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Vehicle motion control of autonomous vehicles using model predictive control

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1 Introduction

The autonomous functions of vehicles can contribute to a safer and more comfortable traffic by reducing the risk and severity of road accidents. In both industry and academic fields, great emphasis is placed on the research and development of advanced driver assistance systems (ADAS), which contribute to the realization of different levels of self-driving.

The justification for driver assistance systems is underlined by the fact that most road accidents are caused by human error. Driver assistance systems can automate certain driving actions, adapt to unexpected obstacles and correct driver errors, making driving safer and more enjoyable. Depending on the autonomous driving level, the system can warn the driver of potential hazards, prepare and activate safety devices and, if necessary, take control of the vehicle to avoid collisions.

On the other hand, higher-level self-driving functions are no longer only activated in case of danger of an accident, but are able to control the vehicle motion by simultaneously controlling the whole powertrain and steering system of the vehicle.

The implementation of these autonomous functions can be divided into two main parts: perception and control. In the perception phase, we need to map the environment of the vehicle using a fusion of different sensors, cameras and radars, on the basis of which the desired vehicle trajectory can be determined. After that we need a control unit that determines the necessary interventions for keeping the vehicle on track based on the signals obtained from perception. Both processes are highly resource intensive, however the continuous increase in computing capacity available for solving engineering problems enables us to design and operate more complex systems, which can be applied sufficiently robustly even in real conditions.

The aim of this study is to construct a complex vehicle model accurate enough to implement autonomous functions for vehicle motion control and to construct an algorithm for vehicle motion control using a model predictive controller (MPC). Finally, the functioning of the plant model and the controller presented in this study are tested in a simulation environment.

1.1 Vehicle motion control

In Chapter 2 of the study we construct a linear complex plant model starting from a simple single track bicycle model. The linearization of the nonlinear state space equations is necessary for the controller to run sufficiently fast, as the MPC controller is computationally demanding. We complete our model with the so called brush tire model, which takes tire deformation also into consideration. In addition, the vehicle model features active or passive rear wheel steering, which models the relationship between the front and rear wheel steering angle. The complete plant model is also supplemented with the dynamics of front and rear wheel steering systems. From the point of view of control, it is essential

that our vehicle model is then connected to the environment (more precisely to the desired path), for this reason we supplement the model with the lateral deviation and heading angle difference from the desired path. Finally, in order to eliminate the steady state error, we also add an integral action to the state feedback. The state space equations are linearized in all cases, we use a LTI model of the plant at every time step. By connecting the above models, we obtain the linear state-space model of our entire plant model, which contains the state-space variables we want to control.

In Chapter 3, we present the longitudinal state-space model in a nutshell.

In Chapter 4, we briefly describe the main steps and methods of trajectory generation, which generates the reference values for the controller.

In Chapter 5, the optimal controller design is presented, and we derive the equations which realize the model predictive control.

Last, but not least, in Chapter 6 we demonstrate some numerical simulations in the CarMaker simulation environment to demonstrate the controller performance.

1.2 Motivation

The main goal of this study is to briefly present the main steps of the process of designing an autonomous controller algorithm. The study is not intended to fully cover all design steps, however, it can be a good summary to introduce the subtasks and the complexity of the problem.

The construction of a complex vehicle dynamics model gets a prominent role, with the help of which the vehicle dynamics can be characterized with a linear state space model. The derivation of the basic ideas and equations of the MPC controller is also emphasized, as this method can be particularly suitable for autonomous functions due to its predictive approach to the problem. Namely, the combination of a linear system model and a quadratic cost function is advantageous in terms of processing time, which is crucial in autonomous functions.

2 Mechanical model

In order to control a vehicle, we need to know its dynamic characteristics. One of the vital steps for this is the construction of the vehicle dynamics model [1, 2, 3], which describes the behavior of the vehicle with sufficient accuracy. In order to minimize the often critical calculation requirement, it is expedient to choose the simplest possible model, which satisfies the requirements we impose on the system. It means when we choose a model, we need to find a balance between complexity and necessity. In addition to the fact that more complex vehicle and tire models require higher calculation requirements, they can often contain empirical parameters. For this reason, the derivation of closed-form analytical solutions is not always possible, and the effect of individual parameters and modeling steps on the entire system are more difficult to detect.

2.1 Single track vehicle model

The most frequently used vehicle model is the single-track model (also known as bicycle model), which is usually sufficient for describing the lateral and yaw motions of the vehicle. This model can be used well from the point of view of lateral control, because we can say with a good approximation that lateral and longitudinal stability can be managed independently. The model can be extended with different tire models that discuss the interaction between tire and road in different depths. The simpler model allows us to derive the closed-form, analytical equations of motion, by which it is capable of characterizing the dynamics of the vehicle sufficiently accurately for the control purposes.

In the single track vehicle model the vehicle is shrunk to its longitudinal plane of symmetry, thereby eliminating the width of the vehicle.

In the model we take the tire deformation also into consideration, the side slip angle is not neglected.

In addition to the characteristics of the deforming tire, the following modeling assumptions are used when we derive the equations of motion. For the forces and moments acting on the vehicle, we take air resistance and the angle of elevation of the road into account. We also assume that the rear wheel is also steered either actively or passively, the effect of which on lateral maneuvers can therefore be investigated later with simulations. Active rear wheel steering means that there is a separate actuator for steering the rear wheel, and passive rear wheel steering suggests that the front and rear steering angle are not independent from each other due to the flexible wheel suspension components.

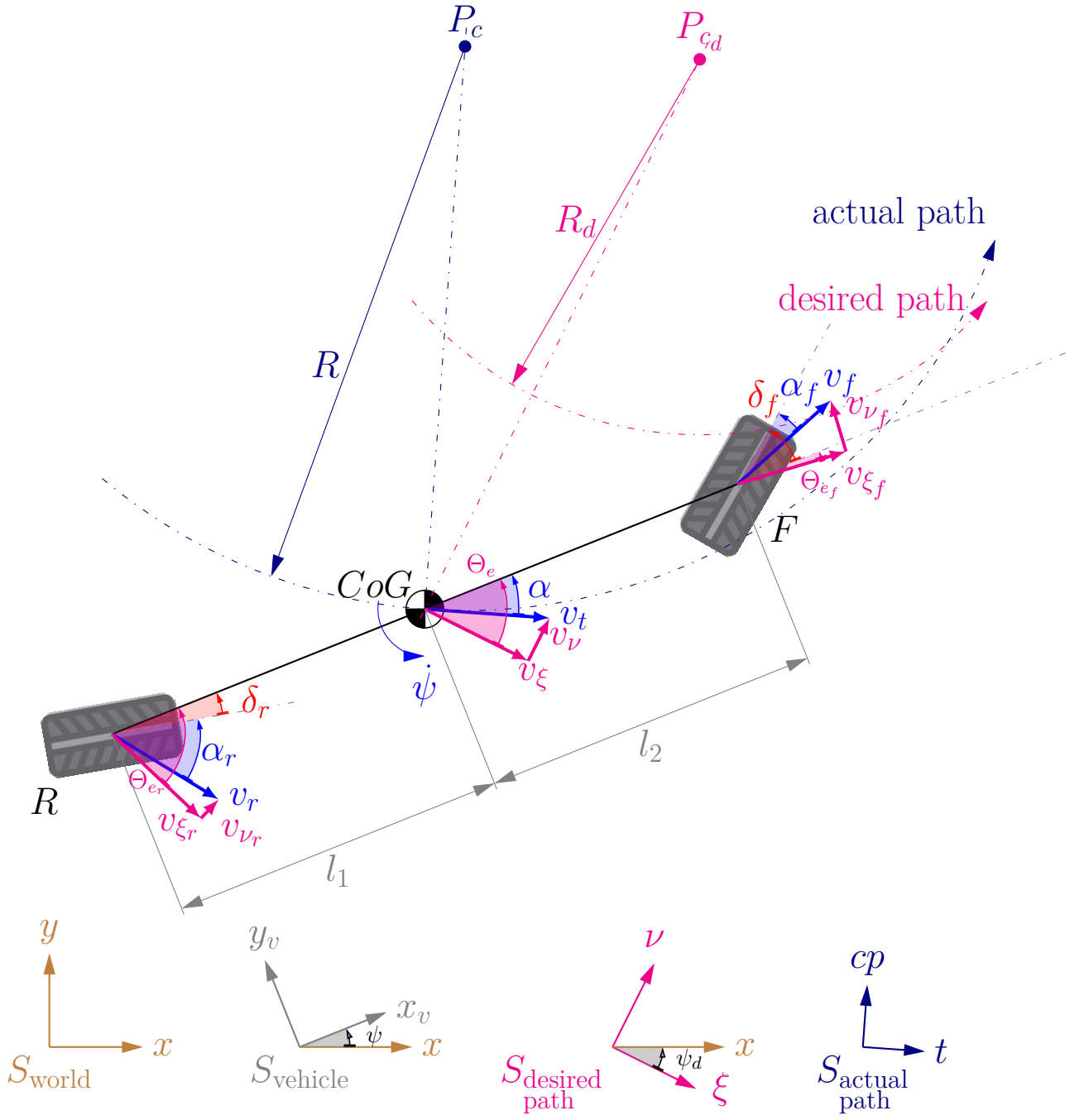


Figure 2.1: Single track model of the vehicle with the coordinate systems, vectors and angles that describe the kinematics of the vehicle.

Based on Fig. 2.1 of the single track model, we define the coordinate systems, vectors and angles to describe the motion of the vehicle. These are the following:

- $S_{vehicle}$ with the (x, y, z) coordinates is the vehicle aligned coordinate system, which has its origin in the CoG of the vehicle. The x -axis is parallel to the ground and forwards, and is in the vehicle's longitudinal plane of symmetry, and the y -axis is perpendicular to the vehicle's longitudinal plane of symmetry and points to the left with the z axis pointing upward, according to the intermediate coordinate system defined by ISO 8855 [4] standard.
- S_{world} with the (X, Y, Z) coordinates is the reference coordinate system. This reference frame is redefined at every moment of time so that it coincides with the vehicle aligned coordinate system $S_{vehicle}$.
- ψ is the yaw angle, with which the vehicle coordinate system $S_{vehicle}$ is rotated about the Z -axis of the S_{world} reference frame. We define S_{world} so that $\psi = 0$, which means that the direction of the two coordinate systems coincides at every time step. However, the yaw angle is needed to describe the correct equations of motion of the vehicle, as $\dot{\psi} \neq 0$.

In fact, we define the reference frame only to be able to define the yaw angle, which is needed to introduce the yaw rate (which, unlike the yaw angle, can be measured, and because of that it will be a state variable of the system).

- P_i and P_{i_d} are the current and desired instant centers of rotation of the entire vehicle body.
- P_c and P_{c_d} are the current and desired centers of the path curvature on which the CoG of the vehicle travels or should travel. Other points of the vehicle have different current and desired center of the path curvature.
- Θ_e is the angle difference between the heading angle of the vehicle and that of the desired path.
- v_e is the vehicle velocity in the direction specified by tangent of the desired path.
- $v_\nu = \dot{\nu}_e$ is the time derivative of the lateral deviation of the vehicle from the desired path. v_ν and Θ_e could be defined at arbitrary point of the vehicle, we assume that they are measured at the CoG.
- α is the side slip angle, which is the signed angle between the vehicle velocity vector at the CoG v_t and the x -axis of the vehicle coordinate system caused by the tire deformation.
- δ is the angle difference between the heading angle of a wheel and the direction of the axis of the single track model.

The above mentioned signed angles and vectors can be defined similarly for the front and rear wheels as well, denoted by the f and r indexes respectively.

2.2 Kinematics of the CoG

Based on the introduced variables we see that the vehicle rotates about Z -axis of the reference frame (or any other fixed point of the reference frame) with an angular speed of $\dot{\psi}$, but on the other hand the CoG of the vehicle rotates about the point P_c with the angular speed of $(\dot{\psi} + \dot{\alpha})$. The two center points P_i and P_c only overlap each other, when $\dot{\alpha} = 0$.

We derive the equations of motion of the vehicle in the road aligned coordinate system S_{world} , which is an inertial reference frame. The vehicle coordinate system $S_{vehicle}$ is moving relatively to this. The relationship between the time derivatives of physical vector quantities in the different (reference and vehicle) coordinate systems is the following:

$$\frac{dA}{dt} = \frac{d_v A}{dt} + \omega \times A, \quad (2.1)$$

where A is the physical vector quantity whose time derivative $\frac{dA}{dt}$ we want to determine in the road aligned (reference) coordinate system as well as $\frac{d_v A}{dt}$ is the time derivative of A in the vehicle coordinate system. ω is the angular speed of the vehicle coordinate system relatively to the reference system, in our case the z component of ω is $\omega_z = (\dot{\psi} + \dot{\alpha})$.

We assume that the path curve the vehicle is traveling on is described by a 2D arc-length parameterized function $\mathcal{G}(s(t))$, where $s = s(t)$ is the arc length, which is also a function of time

$$s(t) = \int_0^t V(\tau) d\tau, \quad (2.2)$$

$$\frac{ds}{dt} = V, \quad (2.3)$$

where $V = V(t)$ is the tangential velocity of the vehicle traveling on the path curve $\mathcal{G}(s(t))$. Because of the assumed 2D nature of $\mathcal{G}(s(t))$ the torsion of this curve is zero, and the curvature of the path is also an arc-length parameterized function $\kappa = \kappa(s(t))$.

2.2.1 Position vector

As we declared above, we assume that the origins of both the road aligned reference frame and that of the vehicle frame coincide. Therefore the position vector of the vehicle CoG within the reference frame is

$$r_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (2.4)$$

2.2.2 Velocity vector

Based on Fig. 2.1, the velocity vector of the CoG yields

$$v_0 = \frac{dr_0}{dt} = \begin{bmatrix} V \cos(\psi + \alpha) \\ V \sin(\psi + \alpha) \\ 0 \end{bmatrix} \approx \begin{bmatrix} V \\ V\alpha \\ 0 \end{bmatrix}, \quad (2.5)$$

where $V = V(t)$ is the speed of the vehicle, $\psi = \psi(t) = 0$ is the yaw angle of the vehicle body around vertical z -axis of the reference frame, and $\alpha = \alpha(t)$ is the side slip angle. Whereas under normal driving conditions the side slip angle α is small ($\alpha < 5^\circ = 0.087$ rad), the small angle approximation can be used for the trigonometric functions:

$$\begin{aligned} \sin(\alpha) &\approx \alpha, \\ \cos(\alpha) &\approx 1. \end{aligned} \quad (2.6)$$

In the following, we will use this small angle approximation for the linearization of the equations. However, it is very important to note that this approximation is only valid if the side slip angle α is sufficiently small, otherwise it may lead to a large model inaccuracy.

2.2.3 Acceleration vector

We determine the acceleration vector at the CoG as the time derivative of the vehicle velocity vector Eq. (2.5), and then we linearize it using the small angle approximation Eq. (2.6) for the side slip angle α :

$$\begin{aligned} a_0 = \dot{v}_0 &= \begin{bmatrix} \dot{V} \cos(\psi + \alpha) - V(\dot{\psi} + \dot{\alpha}) \sin(\psi + \alpha) \\ \dot{V} \sin(\psi + \alpha) + V(\dot{\psi} + \dot{\alpha}) \cos(\psi + \alpha) \\ 0 \end{bmatrix} \stackrel{\psi=0}{=} \\ &\stackrel{\psi=0}{=} \begin{bmatrix} \dot{V} \cos(\alpha) - V(\dot{\psi} + \dot{\alpha}) \sin(\alpha) \\ \dot{V} \sin(\alpha) + V(\dot{\psi} + \dot{\alpha}) \cos(\alpha) \\ 0 \end{bmatrix} \approx \begin{bmatrix} \dot{V} - V(\dot{\psi} + \dot{\alpha})\alpha \\ \dot{V}\alpha + V(\dot{\psi} + \dot{\alpha}) \\ 0 \end{bmatrix}. \end{aligned} \quad (2.7)$$

2.3 Kinematics of the front axle

In this section we derive the motion equations at the front axle, which means the center point of the front axle in our model. These equations will be useful for both determining the side slip angle of the vehicle at the front axle and at planning the state trajectories.

2.3.1 Position vector

The position vector of the front axle in the coordinate system of the vehicle is the constant vector

$$r_f^* = \begin{bmatrix} l_1 \\ 0 \\ 0 \end{bmatrix}. \quad (2.8)$$

The position vector of the front axle in the reference coordinate frame yields

$$r_f = r_0 + r_f^* = \begin{bmatrix} l_1 \\ 0 \\ 0 \end{bmatrix}, \quad (2.9)$$

as we know that the origin of both the vehicle coordinate system and the road-aligned reference coordinate system are located at the CoG of the vehicle.

2.3.2 Velocity vector

The velocity vector of the front wheel is the sum of the velocity of the CoG and the time derivative of the position vector of the front axle in the road-aligned reference coordinate system. To determine the time derivative of the position vector r_f in the reference frame we apply the expression (2.1). By this the velocity vector of the front axle yields

$$\begin{aligned} v_f^{(1)} &= \frac{dr_f}{dt} = \frac{dr_0}{dt} + \frac{dr_f^*}{dt} = v_0 + \underbrace{\frac{d_v r_f^*}{dt}}_0 + \omega \times r_f^* = \begin{bmatrix} V \cos(\psi + \alpha) \\ V \sin(\psi + \alpha) \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} \times \begin{bmatrix} l_1 \\ 0 \\ 0 \end{bmatrix} = \\ &= \begin{bmatrix} V \cos(\psi + \alpha) \\ V \sin(\psi + \alpha) + l_1 \dot{\psi} \\ 0 \end{bmatrix} \stackrel{\psi=0}{=} \begin{bmatrix} V \cos(\alpha) \\ V \sin(\alpha) + l_1 \dot{\psi} \\ 0 \end{bmatrix} \approx \begin{bmatrix} V \\ V\alpha + l_1 \dot{\psi} \\ 0 \end{bmatrix}, \end{aligned} \quad (2.10)$$

where $\omega = \dot{\psi}$ is the angular velocity vector of the vehicle coordinate system relative to the world aligned reference frame and r_f^* is the position vector of the front axle in the vehicle frame.

Based on Fig. 2.1 the front wheel velocity can also be derived as

$$v_f^{(2)} = \begin{bmatrix} V_f \cos(\psi + \delta_f + \alpha_f) \\ V_f \sin(\psi + \delta_f + \alpha_f) \\ 0 \end{bmatrix} \stackrel{\psi=0}{=} \begin{bmatrix} V_f \cos(\delta_f + \alpha_f) \\ V_f \sin(\delta_f + \alpha_f) \\ 0 \end{bmatrix} \approx \begin{bmatrix} V \\ V(\delta_f + \alpha_f) \\ 0 \end{bmatrix}, \quad (2.11)$$

where $V_f \approx V$ is the speed of the front axle, δ_f is the steering angle of the front wheel and α_f is the side slip angle of the front axle.

2.3.3 Side slip angle

The expressions $v_f^{(1)}$ and $v_f^{(2)}$ must be equal. From that equation, if we divide $v_{f_y}^{(1)}$ by $v_{f_x}^{(1)}$ and $v_{f_y}^{(2)}$ by $v_{f_x}^{(2)}$ we get

$$\tan(\delta_f + \alpha_f) = \tan(\alpha) + l_1 \frac{\dot{\psi}}{V \cos(\alpha)}. \quad (2.12)$$

Expressing the side slip angle of the front wheel α_f

$$\alpha_f = \underbrace{\arctan\left(\tan(\alpha) + l_1 \frac{\dot{\psi}}{V \cos(\alpha)}\right)}_{\mathcal{L}_1} - \delta_f = \mathcal{L}_1 - \delta_f. \quad (2.13)$$

If the angles α and $\delta_f + \alpha_f$ are both small, we can apply the small angle approximation

$$\delta_f + \alpha_f \approx \alpha + l_1 \frac{\dot{\psi}}{V}, \quad (2.14)$$

from which

$$\alpha_f \approx \alpha + l_1 \frac{\dot{\psi}}{V} - \delta_f. \quad (2.15)$$

2.3.4 Acceleration vector

We derive the acceleration vector of the front axle by differentiating the velocity vector v_f . First, we differentiate expression Eq. (2.10) of $v_f^{(1)}$ with applying Eq. (2.1)

$$\begin{aligned} a_f^{(1)} &= \dot{v}_f^{(1)} = \frac{d}{dt} (v_0 + \omega \times r_f^*) = a_0 + \dot{\omega} \times r_f^* + \omega \times (\omega \times r_f^*) = \\ &= \begin{bmatrix} \dot{V} \cos(\psi + \alpha) - V \sin(\psi + \alpha) (\dot{\psi} + \dot{\alpha}) - l_1 \dot{\psi}^2 \\ \dot{V} \sin(\psi + \alpha) + V \cos(\psi + \alpha) (\dot{\psi} + \dot{\alpha}) + l_1 \ddot{\psi} \\ 0 \end{bmatrix} \stackrel{\psi=0}{=} \\ &\stackrel{\psi=0}{=} \begin{bmatrix} \dot{V} \cos(\alpha) - V \sin(\alpha) (\dot{\psi} + \dot{\alpha}) - l_1 \dot{\psi}^2 \\ \dot{V} \sin(\alpha) + V \cos(\alpha) (\dot{\psi} + \dot{\alpha}) + l_1 \ddot{\psi} \\ 0 \end{bmatrix} \approx \begin{bmatrix} \dot{V} - V \alpha (\dot{\psi} + \dot{\alpha}) - l_1 \dot{\psi}^2 \\ \dot{V} \alpha + V (\dot{\psi} + \dot{\alpha}) + l_1 \ddot{\psi} \\ 0 \end{bmatrix}. \end{aligned} \quad (2.16)$$

Now we differentiate expression Eq. (2.11) of $v_f^{(2)}$

$$\begin{aligned}
a_f^{(2)} = \dot{v}_f^{(2)} &= \frac{d}{dt} \left(\begin{bmatrix} V_f \cos(\psi + \delta_f + \alpha_f) \\ V_f \sin(\psi + \delta_f + \alpha_f) \\ 0 \end{bmatrix} \right) = \\
&= \begin{bmatrix} \dot{V}_f \cos(\psi + \delta_f + \alpha_f) - V_f \sin(\psi + \delta_f + \alpha_f) (\dot{\psi} + \dot{\delta}_f + \dot{\alpha}_f) \\ \dot{V}_f \sin(\psi + \delta_f + \alpha_f) + V_f \cos(\psi + \delta_f + \alpha_f) (\dot{\psi} + \dot{\delta}_f + \dot{\alpha}_f) \\ 0 \end{bmatrix} \stackrel{\psi=0}{=} \\
&\begin{bmatrix} \dot{V}_f \cos(\delta_f + \alpha_f) - V_f \sin(\delta_f + \alpha_f) (\dot{\psi} + \dot{\delta}_f + \dot{\alpha}_f) \\ \dot{V}_f \sin(\delta_f + \alpha_f) + V_f \cos(\delta_f + \alpha_f) (\dot{\psi} + \dot{\delta}_f + \dot{\alpha}_f) \\ 0 \end{bmatrix} \approx \begin{bmatrix} \dot{V} - V(\delta_f + \alpha_f) (\dot{\psi} + \dot{\delta}_f + \dot{\alpha}_f) \\ \dot{V}(\delta_f + \alpha_f) + V(\dot{\psi} + \dot{\delta}_f + \dot{\alpha}_f) \\ 0 \end{bmatrix}, \tag{2.17}
\end{aligned}$$

where we assumed that $V_f \approx V$.

2.4 Kinematics of the rear axle

In this section we derive the motion equations at the rear axle, which means the center point of the front axle in our model. The derivation is carried out in a completely analogous way, as in the case of the front axle, the only difference is the first coordinate of the position vector. For this reason, we only present the parametric velocity and acceleration reduction formulas and the linearized results, the calculation process is not detailed here.

2.4.1 Position vector

The position vector of the rear axle in the coordinate system of the vehicle is the constant vector

$$r_r^* = \begin{bmatrix} -l_2 \\ 0 \\ 0 \end{bmatrix}. \tag{2.18}$$

The position vector of the rear axle in the reference coordinate frame yields

$$r_r = r_0 + r_r^* = \begin{bmatrix} -l_2 \\ 0 \\ 0 \end{bmatrix}, \tag{2.19}$$

as we know that the origin of both the vehicle coordinate system and the road-aligned reference coordinate system locates at the CoG of the vehicle.

2.4.2 Velocity vector

The velocity vector of the rear axle yields

$$v_r^{(1)} = v_0 + \omega \times r_r^* \approx \begin{bmatrix} V \\ V\alpha - l_2\dot{\psi} \\ 0 \end{bmatrix}, \quad (2.20)$$

or based on Fig. 2.1 we can also express it as

$$v_r^{(2)} = \begin{bmatrix} V_r \cos(\psi + \delta_r + \alpha_r) \\ V_r \sin(\psi + \delta_r + \alpha_r) \\ 0 \end{bmatrix} \stackrel{\psi=0}{=} \begin{bmatrix} V_r \cos(\delta_r + \alpha_r) \\ V_r \sin(\delta_r + \alpha_r) \\ 0 \end{bmatrix} \approx \begin{bmatrix} V \\ V(\delta_r + \alpha_r) \\ 0 \end{bmatrix}. \quad (2.21)$$

2.4.3 Side slip angle

By using the equivalence of $v_r^{(1)}$ and $v_r^{(2)}$, similarly to the front wheel axle the side slip angle of the rear axle can be derived as

$$\alpha_r \approx \alpha - l_2 \frac{\dot{\psi}}{V} - \delta_r. \quad (2.22)$$

2.4.4 Acceleration vector

By differentiating $v_r^{(1)}$ we obtain the acceleration vector of the rear axle

$$a_r^{(1)} = \dot{v}_r^{(1)} = a_0 + \dot{\omega} \times r_r^* + \omega \times (\omega \times r_r^*) \approx \begin{bmatrix} \dot{V} - V\alpha(\dot{\psi} + \dot{\alpha}) + l_2\dot{\psi}^2 \\ \dot{V}\alpha + V(\dot{\psi} + \dot{\alpha}) - l_2\ddot{\psi} \\ 0 \end{bmatrix}, \quad (2.23)$$

or by differentiating $v_r^{(2)}$ we get

$$a_r^{(2)} = \dot{v}_r^{(2)} \approx \begin{bmatrix} \dot{V} - V(\delta_r + \alpha_r)(\dot{\psi} + \dot{\delta}_r + \dot{\alpha}_r) \\ \dot{V}(\delta_r + \alpha_r) + V(\dot{\psi} + \dot{\delta}_r + \dot{\alpha}_r) \\ 0 \end{bmatrix}. \quad (2.24)$$

2.5 Kinetic description of the motion of the vehicle

To determine the equations of motion characterizing the dynamics of the vehicle, we first write down the force and moment equations for the CoG based on Newton's second law of motion:

$$\sum F = F_f + F_r + F_{ad} + F_g = ma, \quad (2.25)$$

$$\sum M_{CoG} = M_f + M_r + M_{ad} = \dot{\Pi}_{CoG}, \quad (2.26)$$

where

- F_f is the force vector on the contact point (surface) of the front wheel of the single-track vehicle model.
- M_f is the moment vector caused by F_f at the CoG of the single-track vehicle model.
- F_r is the force vector on the contact point (surface) of the rear wheel of the single-track vehicle model.
- M_r is the moment vector caused by F_r at the CoG of the single-track vehicle model.
- F_{ad} is the force vector of the air drag.
- M_{ad} is the moment vector caused by F_{ad} at the CoG of the single-track vehicle model.
- F_g is the gravitational force vector at the CoG.

The longitudinal and vertical forces acting on the vehicle can be seen in Fig. 2.2, the lateral forces are shown in Fig. 2.3.

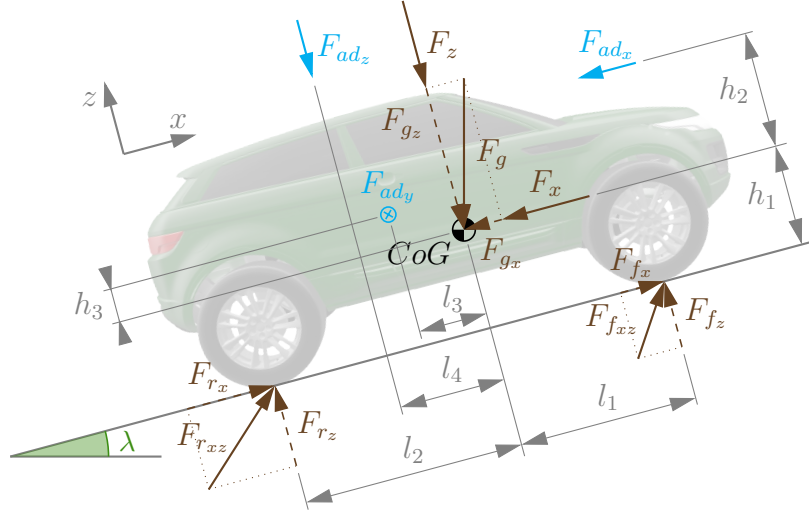


Figure 2.2: Side view of the vehicle with the longitudinal and the vertical forces.

2.5.1 Resultant force and moment

Based on the acceleration vector Eq. (2.7) defined for the CoG, the resultant force is given as follows, using the linearization Eq. (2.6):

$$\sum F = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = ma = \begin{bmatrix} m\dot{V} \cos(\alpha) - mV(\dot{\psi} + \dot{\alpha}) \sin(\alpha) \\ m\dot{V} \sin(\alpha) + mV(\dot{\psi} + \dot{\alpha}) \cos(\alpha) \\ 0 \end{bmatrix} \approx \begin{bmatrix} m\dot{V} - mV(\dot{\psi} + \dot{\alpha})\alpha \\ m\dot{V}\alpha + mV(\dot{\psi} + \dot{\alpha}) \\ 0 \end{bmatrix}. \quad (2.27)$$

The moment of inertia yields

$$\sum M_{CoG} = \begin{bmatrix} M_x \\ M_y \\ M_z \end{bmatrix} = \dot{\Pi}_{CoG} = J_{CoG}\dot{\omega} + \underbrace{\omega \times (J_{CoG}\omega)}_{=0} = \underbrace{\begin{bmatrix} J_{xx} & J_{xy} & J_{zx} \\ J_{yx} & J_{yy} & J_{yx} \\ J_{zx} & J_{zy} & J_{zz} \end{bmatrix}}_{J_{CoG}} \underbrace{\begin{bmatrix} 0 \\ 0 \\ \ddot{\psi} \end{bmatrix}}_{\ddot{\omega}} = \begin{bmatrix} J_{xz}\ddot{\psi} \\ J_{yz}\ddot{\psi} \\ J_{zz}\ddot{\psi} \end{bmatrix}, \quad (2.28)$$

where $\omega = \dot{\psi}$ is the angular velocity of the vehicle coordinate system relative to the reference coordinate system, and J_{CoG} is the matrix of the vehicle inertia tensor at the CoG. The additional term due to spatial motion $\omega \times \Pi_{CoG} = \omega \times (J_{CoG}\omega)$ is going to be 0, as ω has a nonzero component only in the Z direction. It is important to mention here that the pitch and roll angles of the vehicle are not modeled, in the track-bound coordinate system we examine only planar motion (even in the case of a sloping road).

2.5.2 Aerodynamic forces and moments

The air drag force can be expressed as

$$F_{ad} = \begin{bmatrix} F_{adx} \\ F_{ady} \\ F_{adz} \end{bmatrix} = \begin{bmatrix} \frac{\rho}{2}c_D Av^2 \\ \frac{\rho}{2}c_S Av^2 \\ \frac{\rho}{2}c_L Av^2 \end{bmatrix}, \quad (2.29)$$

from which the air drag torque vector yields

$$M_{ad} = \begin{bmatrix} M_{adx} \\ M_{ady} \\ M_{adz} \end{bmatrix} = \begin{bmatrix} \frac{\rho}{2}c_{rm} l Av^2 \\ \frac{\rho}{2}c_{pm} l Av^2 \\ \frac{\rho}{2}c_{ym} l Av^2 \end{bmatrix}, \quad (2.30)$$

where ρ is the air density, c_D , c_S , c_L are the drag, the side and the lift coefficients, respectively. A is the reference area of the vehicle, l is the reference length, v is the relative wind speed, c_{rm} , c_{pm} , c_{ym} are the roll, pitch and yaw moment coefficients, respectively. All these parameters (except from air density) are provided by the manufacturer. The actual values of the coefficients are highly influenced by the relative direction of the wind speed vector. Air drag forces and moments are considered as disturbances for the vehicle to be controlled.

2.5.3 Gravitational forces

The gravitational force vector yields

$$F_g = \begin{bmatrix} F_{gx} \\ F_{gy} \\ F_{gz} \end{bmatrix}. \quad (2.31)$$

As this force acts at the CoG, its moment on the CoG is zero:

$$M_g = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (2.32)$$

2.5.4 Front wheel forces and moments

In this section we determine the forces underneath the front wheel acting on the vehicle and the moments caused by these forces. According to the free body diagram in Fig. 2.3 the force vector at the contact point between the front tire and the road surface yields

$$F_f = \begin{bmatrix} \cos(\delta_f) & -\sin(\delta_f) & 0 \\ \sin(\delta_f) & \cos(\delta_f) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} F_{f_l} \\ F_{f_s} \\ F_{f_z} \end{bmatrix} = \begin{bmatrix} F_{f_l} \cos(\delta_f) - F_{f_s} \sin(\delta_f) \\ F_{f_l} \sin(\delta_f) + F_{f_s} \cos(\delta_f) \\ F_{f_z} \end{bmatrix} \approx \begin{bmatrix} F_{f_l} - F_{f_s} \delta_f \\ F_{f_l} \delta_f + F_{f_s} \\ F_{f_z} \end{bmatrix}, \quad (2.33)$$

where F_{f_l} is the longitudinal, F_{f_s} is the lateral and F_{f_z} is the vertical force component. We used the fact that the front wheel is rotated by the steering angle δ_f with respect to the z -axis of the vehicle, and we assumed that this angle is small enough so that the small angle approximation can be applied.

In reality the force vector F_f does not act exactly under the wheel center, the pneumatic trail caused by the tire deformation must be taken into account [5, 6]. Due to the shape of the lateral load distribution on the contact patch between the tire and the road surface the resultant lateral force acts at a small distance behind the wheel center. This distance is called the pneumatic trail, which is typically in the range of $[0, 0.1]$ m. We take this effect into account when determining the moment vector M_f acting on the CoG of the vehicle, caused by the force vector F_f :

$$\begin{aligned} M_f &= r_f^* \times F_f = \\ &= \begin{bmatrix} l_1 \\ 0 \\ -h_1 \end{bmatrix} \times \begin{bmatrix} F_{f_l} \cos(\delta_f) - F_{f_s} \sin(\delta_f) \\ 0 \\ F_{f_z} \end{bmatrix} + \begin{bmatrix} l_1 - t_f \\ 0 \\ -h_1 \end{bmatrix} \times \begin{bmatrix} 0 \\ F_{f_l} \sin(\delta_f) + F_{f_s} \cos(\delta_f) \\ 0 \end{bmatrix} = \\ &= \begin{bmatrix} (F_{f_l} \sin(\delta_f) + F_{f_s} \cos(\delta_f)) h_1 \\ -F_{f_z} l_1 - (F_{f_l} \cos(\delta_f) - F_{f_s} \sin(\delta_f)) h_1 \\ (F_{f_l} \sin(\delta_f) + F_{f_s} \cos(\delta_f)) (l_1 - t_f) \end{bmatrix} \approx \begin{bmatrix} (F_{f_l} \delta_f + F_{f_s}) h_1 \\ -F_{f_z} l_1 - (F_{f_l} - F_{f_s} \delta_f) h_1 \\ (F_{f_l} \delta_f + F_{f_s}) (l_1 - t_f) \end{bmatrix}, \end{aligned} \quad (2.34)$$

where h_1 is the height of the CoG from the road surface, l_1 is the distance of the front wheel from the CoG in x -direction and t_f is the pneumatic trail at the front wheel. Because of the pneumatic trail, we decoupled the y -component of the front wheel force from the other components.

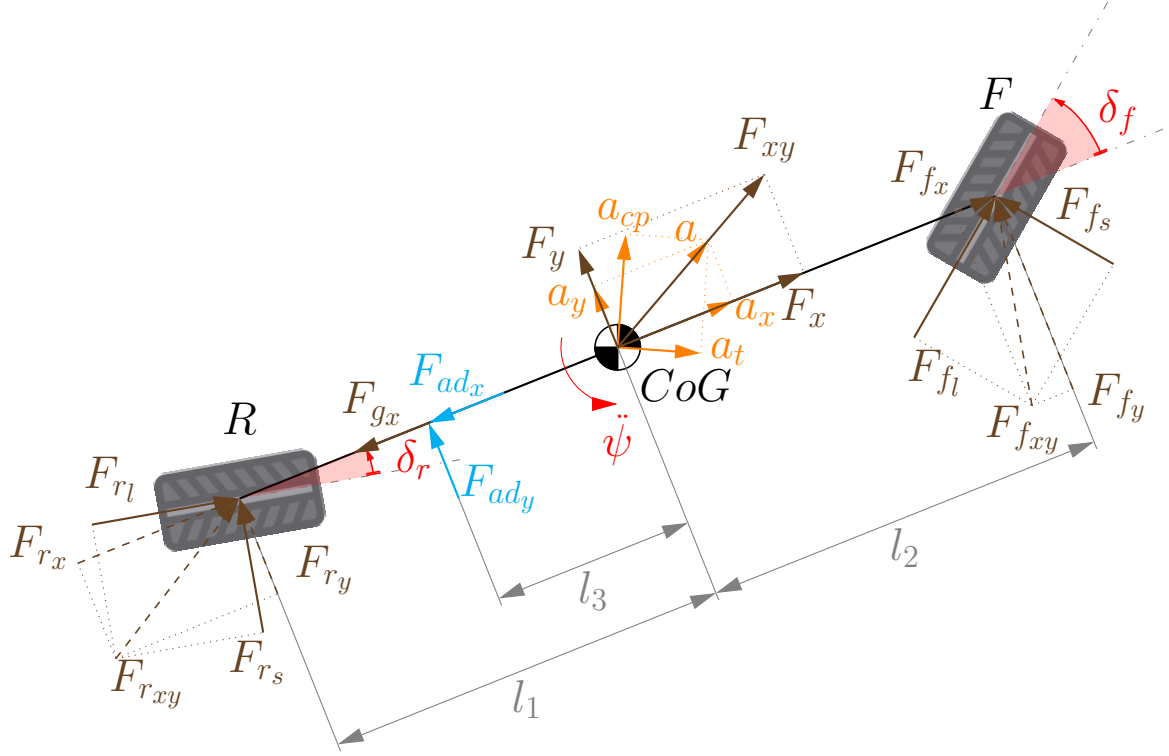


Figure 2.3: Free body diagram of the vehicle with the longitudinal and lateral forces.

2.5.4.1 Brush tire model

It is known that the steady-state lateral force under the front wheel can be expressed as linear function of α_f

$$F_{f_s}^s(\alpha_f) = c_f \alpha_f, \quad (2.35)$$

where

$$c_f = \left. \frac{\partial F_{f_s}^S}{\partial \alpha_f} \right|_{\alpha_f=0}, \quad (2.36)$$

is the cornering stiffness and α_f is the side slip angle of the front axle. Since this expression is only valid in steady-state case, we introduce a dynamic model to improve the model accuracy. Moreover, we take the relaxation effect of the tire into consideration, as the side slip angle of the front axle and the front tire are different. To deal with this phenomenon, we assume that the tire carcass is not rigid, and we model it with the so called brush tire model [5], which is based on the Kelvin-Voigt material model. It uses a parallel connected viscous damper and an elastic spring to describe the viscoelastic material behavior of the tire. We approximate the relation between the side-slip angle of the front tire and that of the front axle by a first order differential equation [7]

$$\frac{\sigma_{f_y}}{V_{f_x}} \dot{\tilde{\alpha}}_f + \tilde{\alpha}_f = \alpha_f, \quad (2.37)$$

where σ_{f_y} is the relaxation length of the front tire, V_{f_x} is the longitudinal wheel speed, $\tilde{\alpha}_f$ is the side slip angle of the front tire and α_f is the side slip angle of the front axle. It

can be shown that for the relaxation length it applies

$$\sigma_{f_y} = \frac{c_f}{c_y}, \quad (2.38)$$

where

$$c_y = \left. \frac{\partial F_{f_s}}{\partial y} \right|_{y=0}, \quad (2.39)$$

is the lateral spring stiffness (carcass stiffness) of the front tire. We assume that this parameter is independent from the tire loads and the front and rear tires are of the same type. Because of that, we use the same parameter c_y for both the front and rear axles. By substituting Eq. (2.38) into Eq. (2.37), and then multiplying it by c_f and using Eq. (2.35) we get that

$$\frac{c_f}{c_y V_{f_x}} c_f \dot{\alpha}_f + c_f \tilde{\alpha}_f = c_f \alpha_f, \quad (2.40)$$

$$\frac{c_f}{c_y V_{f_x}} \dot{F}_{f_s}^D + F_{f_s}^D = F_{f_s}^S, \quad (2.41)$$

where $F_{f_s}^D$ is the dynamic and $F_{f_s}^S$ is the steady-state lateral force under the front wheel ($\lim_{t \rightarrow \infty} F_{f_s}^D = F_{f_s}^S$). As $F_{f_s}^D$ is the real force acting on the front tire and $F_{f_s}^S$ is merely a fictive force, we will use Eq. (2.41) in the following form

$$\frac{c_f}{c_y V_{f_x}} \dot{F}_{f_s} + F_{f_s} = c_f \alpha_f. \quad (2.42)$$

Let the relaxation time constant be defined as

$$\tau_f = \frac{c_f}{c_y V_{f_x}} \approx \frac{c_f}{c_y V}. \quad (2.43)$$

From that the simple first order form of the lateral force underneath the front tire is

$$\tau_f \dot{F}_{f_s} + F_{f_s} = c_f \alpha_f. \quad (2.44)$$

We also know that the relaxation time constant multiplied by the longitudinal wheel speed is the relaxation length

$$\sigma_f = V_{f_x} \tau_f. \quad (2.45)$$

Rearranging Eq. (2.44) and substituting expression Eq. (2.13) for α_f as well as Eq. (2.43) for τ_f into it, we obtain

$$\dot{F}_{f_s} = \frac{1}{\tau_f} (c_f \alpha_f - F_{f_s}) = \frac{c_f}{\tau_f} (\mathcal{L}_1 - \delta_f) - \frac{1}{\tau_f} F_{f_s} = V c_y (\mathcal{L}_1 - \delta_f) - \frac{V c_y}{c_f} F_{f_s}. \quad (2.46)$$

which, due to the non-linearity of $\mathcal{L}_1$, is a non-linear equation. If we apply the small angle approximation, we rather substitute Eq. (2.15) into Eq. (2.44) and after rearranging we get the following linear equation

$$\dot{F}_{f_s} = \frac{1}{\tau_f} (c_f \alpha_f - F_{f_s}) = \frac{c_f}{\tau_f} \left(\alpha + l_1 \frac{\dot{\psi}}{V} - \delta_f \right) - \frac{1}{\tau_f} F_{f_s}. \quad (2.47)$$

After rearranging it for a more useful form for the state space representation

$$\dot{F}_{f_s} = \frac{c_f l_1}{\tau_f V} \dot{\psi} + \frac{c_f}{\tau_f} \alpha - \frac{1}{\tau_f} F_{f_s} - \frac{c_f}{\tau_f} \delta_f. \quad (2.48)$$

Finally, by substituting Eq. (2.43) into this equation we obtain

$$\dot{F}_{f_s} = c_y l_1 \dot{\psi} + c_y V \alpha - \frac{c_y V}{c_f} F_{f_s} - c_y V \delta_f. \quad (2.49)$$

By this F_{f_s} becomes a promising state candidate for a linear state space representation of the plant model.

From that we make further considerations. Rearranging Eq. (2.44) as follows

$$F_{f_s} = c_f \alpha_f - \tau_f \dot{F}_{f_s}, \quad (2.50)$$

From that it can be seen that if α_f can be measured or estimated and if we assume that the term $\tau_f \dot{F}_{f_s}$ can be modeled with Gaussian noise such that

$$v_{F_{f_s}} = -\tau_f \dot{F}_{f_s}, \quad (2.51)$$

where $v_{F_{f_s}}$ is Gaussian noise with covariance matrix $R_{F_{f_s}} \delta(t - \tau)$. Then the measured value of the lateral force underneath the front wheels is

$$F_{f_s} = c_f \alpha_f + v_{F_{f_s}}. \quad (2.52)$$

If, instead of α_f , only α and δ_f can be measured or estimated, then substituting Eq. (2.15) into the above equations yields

$$F_{f_s} = c_f \left(\alpha + l_1 \frac{\dot{\psi}}{V} - \delta_f \right) + v_{F_{f_s}}. \quad (2.53)$$

This last expression may then be applied for the output equation of the vehicle state space representation.

2.5.5 Rear wheel forces and moments

Completely analogous to the equations written for the front wheels, the forces acting on the rear wheels can also be determined with the same formulas, the only difference is in the position vector. For this reason, we do not detail the calculations here. The force vector at the contact point between the rear tire and the road surface yields

$$F_r = \begin{bmatrix} F_{r_l} \cos(\delta_r) - F_{r_s} \sin(\delta_r) \\ F_{r_l} \sin(\delta_r) + F_{r_s} \cos(\delta_r) \\ F_{r_z} \end{bmatrix} \approx \begin{bmatrix} F_{r_l} - F_{r_s} \delta_r \\ F_{r_l} \delta_r + F_{r_s} \\ F_{r_z} \end{bmatrix}, \quad (2.54)$$

The moment vector M_r acting on the CoG of the vehicle, caused by the force vector F_r is

$$M_r = r_r^* \times F_r = \begin{bmatrix} (F_{r_l} \sin(\delta_r) + F_{r_s} \cos(\delta_r)) h_1 \\ F_{r_z} l_2 - (F_{r_l} \cos(\delta_r) - F_{r_s} \sin(\delta_r)) h_1 \\ (F_{r_l} \sin(\delta_r) + F_{r_s} \cos(\delta_r)) (-l_2 - t_r) \end{bmatrix} \approx \begin{bmatrix} (F_{r_l} \delta_r + F_{r_s}) h_1 \\ F_{r_z} l_2 - (F_{r_l} - F_{r_s} \delta_r) h_1 \\ (F_{r_l} \delta_r + F_{r_s}) (-l_2 - t_r) \end{bmatrix}, \quad (2.55)$$

To write the state space equations, we will need the following two equations derived from the brush tire model similarly to the front wheel

$$\dot{F}_{r_s} = -c_y l_2 \dot{\psi} + c_y V \alpha - \frac{c_y V}{c_r} F_{r_s} - c_y V \delta_r, \quad (2.56)$$

$$F_{r_s} = c_r \left(\alpha - l_2 \frac{\dot{\psi}}{V} - \delta_r \right) + v_{F_{r_s}}. \quad (2.57)$$

2.6 Force and moment equilibrium

As we have derived all force vectors, we can calculate the resulting force vector by simply substituting the individual force vectors Eq. (2.27), Eq. (2.29), Eq. (2.31), Eq. (2.33) and Eq. (2.54) into equation Eq. (2.25). The resulting force vector yields

$$\begin{aligned} \sum F = & \underbrace{\begin{bmatrix} F_{f_l} \cos(\delta_f) - F_{f_s} \sin(\delta_f) \\ F_{f_l} \sin(\delta_f) + F_{f_s} \cos(\delta_f) \\ F_{f_z} \end{bmatrix}}_{F_f} + \underbrace{\begin{bmatrix} F_{r_l} \cos(\delta_r) - F_{r_s} \sin(\delta_r) \\ F_{r_l} \sin(\delta_r) + F_{r_s} \cos(\delta_r) \\ F_{r_z} \end{bmatrix}}_{F_r} + \underbrace{\begin{bmatrix} \frac{\rho}{2} c_D A v^2 \\ \frac{\rho}{2} c_S A v^2 \\ \frac{\rho}{2} c_L A v^2 \end{bmatrix}}_{F_{ad}} + \\ & + \underbrace{\begin{bmatrix} F_{g_x} \\ F_{g_y} \\ F_{g_z} \end{bmatrix}}_{F_g} = \begin{bmatrix} m \dot{V} \cos(\alpha) - m V (\dot{\psi} + \dot{\alpha}) \sin(\alpha) \\ m \dot{V} \sin(\alpha) + m V (\dot{\psi} + \dot{\alpha}) \cos(\alpha) \\ 0 \end{bmatrix}. \end{aligned} \quad (2.58)$$

By applying the small angle approximation, the resulting force vector has the form

$$\sum F \approx \underbrace{\begin{bmatrix} F_{f_l} - F_{f_s} \delta_f \\ F_{f_l} \delta_f + F_{f_s} \\ F_{f_z} \end{bmatrix}}_{F_f} + \underbrace{\begin{bmatrix} F_{r_l} - F_{r_s} \delta_r \\ F_{r_l} \delta_r + F_{r_s} \\ F_{r_z} \end{bmatrix}}_{F_r} + \underbrace{\begin{bmatrix} \frac{\rho}{2} c_D A v^2 \\ \frac{\rho}{2} c_S A v^2 \\ \frac{\rho}{2} c_L A v^2 \end{bmatrix}}_{F_{ad}} + \underbrace{\begin{bmatrix} F_{g_x} \\ F_{g_y} \\ F_{g_z} \end{bmatrix}}_{F_g} = \begin{bmatrix} m \dot{V} - m V (\dot{\psi} + \dot{\alpha}) \alpha \\ m \dot{V} \alpha + m V (\dot{\psi} + \dot{\alpha}) \\ 0 \end{bmatrix}. \quad (2.59)$$

Similarly for the moment vectors, if we substitute the individual moment vectors Eq. (2.28), Eq. (2.30), Eq. (2.34) and Eq. (2.55) into equation Eq. (2.26) we obtain

the following for the resulting moment vector for the CoG

$$\begin{aligned} \sum M = & \underbrace{\begin{bmatrix} (F_{f_l} \sin(\delta_f) + F_{f_s} \cos(\delta_f)) h_1 \\ -F_{f_z} l_1 - (F_{f_l} \cos(\delta_f) - F_{f_s} \sin(\delta_f)) h_1 \\ (F_{f_l} \sin(\delta_f) + F_{f_s} \cos(\delta_f)) (l_1 - t_f) \end{bmatrix}}_{M_f} + \\ & + \underbrace{\begin{bmatrix} (F_{r_l} \sin(\delta_r) + F_{r_s} \cos(\delta_r)) h_1 \\ F_{r_z} l_2 - (F_{r_l} \cos(\delta_r) - F_{r_s} \sin(\delta_r)) h_1 \\ (F_{r_l} \sin(\delta_r) + F_{r_s} \cos(\delta_r)) (-l_2 - t_r) \end{bmatrix}}_{M_r} + \underbrace{\begin{bmatrix} \frac{\rho}{2} c_{rm} l A v^2 \\ \frac{\rho}{2} c_{pm} l A v^2 \\ \frac{\rho}{2} c_{ym} l A v^2 \end{bmatrix}}_{M_{ad}} = \begin{bmatrix} J_{xz} \ddot{\psi} \\ J_{yz} \ddot{\psi} \\ J_{zz} \ddot{\psi} \end{bmatrix}. \end{aligned} \quad (2.60)$$

By applying the small angle approximation, the resulting moment vector for the CoG has the form

$$\begin{aligned} \sum M \approx & \underbrace{\begin{bmatrix} (F_{f_l} \delta_f + F_{f_s}) h_1 \\ -F_{f_z} l_1 - (F_{f_l} - F_{f_s} \delta_f) h_1 \\ (F_{f_l} \delta_f + F_{f_s}) (l_1 - t_f) \end{bmatrix}}_{M_f} + \underbrace{\begin{bmatrix} (F_{r_l} \delta_r + F_{r_s}) h_1 \\ F_{r_z} l_2 - (F_{r_l} - F_{r_s} \delta_r) h_1 \\ (F_{r_l} \delta_r + F_{r_s}) (-l_2 - t_r) \end{bmatrix}}_{M_r} + \underbrace{\begin{bmatrix} \frac{\rho}{2} c_{rm} l A v^2 \\ \frac{\rho}{2} c_{pm} l A v^2 \\ \frac{\rho}{2} c_{ym} l A v^2 \end{bmatrix}}_{M_{ad}} = \begin{bmatrix} J_{xz} \ddot{\psi} \\ J_{yz} \ddot{\psi} \\ J_{zz} \ddot{\psi} \end{bmatrix}. \end{aligned} \quad (2.61)$$

2.7 State-space equations

In this section, we express the derivatives of the state candidates from the previously written equations in the form required for the state space representation.

2.7.1 Yaw acceleration

After rearranging the equation Eq. (2.61) of the yaw moment balance in the z direction we obtain the following equation for $\ddot{\psi}$:

$$\ddot{\psi} = \frac{l_1 - t_f}{J_{zz}} F_{f_s} - \frac{l_2 + t_r}{J_{zz}} F_{r_s} + \frac{F_{f_l} l_1}{J_{zz}} \delta_f - \frac{F_{r_l} l_1}{J_{zz}} \delta_r + \frac{M_{adz}}{J_{zz}}. \quad (2.62)$$

2.7.2 Side slip angle derivative

Assuming that we can measure $\ddot{\psi}$, this equation can serve as both state and output equation. From the equation Eq. (2.60) of the lateral force balance we can derive the side slip angular speed $\dot{\alpha}$:

$$\dot{\alpha} = -\dot{\psi} - \frac{\dot{V}}{V} \alpha + \frac{1}{Vm} F_{f_s} + \frac{1}{Vm} F_{r_s} + \frac{F_{f_l}}{Vm} \delta_f + \frac{F_{r_l}}{Vm} \delta_r + \frac{F_{ady} + F_{gy}}{Vm}. \quad (2.63)$$

2.7.3 Lateral acceleration

As the side slip angle α appears here as a state variable, the question arises how we can measure the side slip angle α [8]. It can be done by applying visual odometry functions using optical flow fields detected for instance by the front video camera. Another way is by using the Electric Power Assisted Steering (EPAS), which can measure the so called alignment force on the rack evolved by the lateral forces underneath the front wheels. Knowing the geometry of the wheel suspension and the steering mechanism, the lateral force beneath the front wheels can be indirectly measured this way. Moreover by knowing the cornering stiffness and the relaxation time or the relaxation length of the front tires, the resulting side slip angle α_f can easily be determined from Eq. (2.49). After that, from equation Eq. (2.33) we obtain the side slip angle of the car. Similarly from equation Eq. (2.54) we can derive α_r and F_{r_s} for the rear wheels. This means we can theoretically measure α , α_r , α_f , F_{f_s} and F_{r_s} . If other sensors, such as GPS or LIDAR are also available, the side slip angle and the other quantities can be estimated more accurately by sensor fusion.

The lateral acceleration a_y can be easily measured by the inertial measurement unit (IMU) sensor. Rearranging Eq. (2.58) and using Eq. (2.60) we can express the lateral acceleration at the CoG as follows:

$$a_y = \frac{F_{f_s}}{m} + \frac{F_{r_s}}{m} + \frac{F_{f_l}\delta_f}{m} + \frac{F_{r_l}\delta_r}{m} + \frac{F_{a_d_y} + F_{g_y}}{m}. \quad (2.64)$$

2.8 Rear wheel steering

We assume that the vehicle is also equipped with a rear axle steering module (RASM) besides the EPAS, which means that both axles are actively steered. If there is no active rear wheel steering then we can completely ignore the rear wheel steering or, if the passive rear wheel steering, due to the flexible wheel suspension components, is not negligible, then we may introduce some model for the rear wheel steering [9]. Indeed, in this case the rear wheel steering angle and the front wheel steering angle are not independent from one another. There must be a function between them which is implemented by the mechanics of the rear wheel suspension. We may assume that, in the simplest case, it has the form

$$\delta_r = f(\delta_f, F_r), \quad (2.65)$$

where F_r is the resulting force vector on the contact surface between the rear tire and the road surface. This function can be unknown or too complex including nonlinearity. Because of that we choose a simple but satisfyingly accurate model, assuming a proportional relationship in a small vicinity of the current operating point. The assumed relationship yields:

$$\delta_r = \Xi \delta_f, \quad (2.66)$$

where Ξ is constant and it is determined in each operating point, which are defined by maximum three coordinates: the front steering angle request $\delta_{f_{req}}$, and if there is active

rear wheel steering, then the rear steering angle request δ_{req} , and the longitudinal vehicle speed V .

If we do not want to handle millions of operating points, then we have to apply various simplifications. Such a simplification can be for example the assumption of both steering angles δ_f and δ_r are small. As a consequence the earlier applied small-angle approximation holds as well as the quantities $F_{f_l}\delta_f$ and $F_{r_l}\delta_r$ are mostly small compared to F_{f_s} and F_{r_s} , respectively. This means that we may neglect the effect of $F_{f_l}\delta_f$ and $F_{r_l}\delta_r$ on the lateral dynamics, that is

$$F_{f_l}\delta_f \approx 0, \quad (2.67)$$

$$F_{r_l}\delta_r \approx 0. \quad (2.68)$$

Note that although the front and the rear steering angle requests are coordinates of the operating points, this does not contradict the other assumption of the small steering angles. EPAS and RASM models may need, due to their strange nonlinearity, quite many different parameters even within the small range of $[0^\circ, 1^\circ]$. This is because a front steering angle of 1° means a quite strong steering. Assume a vehicle wheelbase of 2.6 m. Now based on the Ackermann steering geometry we get a steering radius of

$$R = \frac{l}{\tan(\delta_f)} = \frac{2.6 \text{ m}}{\tan(1^\circ)} \approx 150 \text{ m}. \quad (2.69)$$

At European country roads, the permitted highest speed is of 90 km/h and the minimum curve radius is usually not smaller than 300 m. In the USA the minimum curve radii are above the European values. As another simplification we also assume that the vehicle is traveling with about constant speed, that is it neither accelerates nor decelerates intensively.

$$\dot{V} \approx 0. \quad (2.70)$$

2.9 Simplified state-space equations

Applying these simplifications for Eq. (2.62), Eq. (2.63), Eq. (2.64), we obtain the following expressions for the yaw acceleration $\ddot{\psi}$, the side slip angle derivative $\dot{\alpha}$ and the lateral acceleration a_y :

$$\ddot{\psi} = \frac{l_1 - t_f}{J_{zz}} F_{f_s} - \frac{l_2 + t_r}{J_{zz}} F_{r_s} + \frac{M_{adz}}{J_{zz}}, \quad (2.71)$$

$$\dot{\alpha} = -\dot{\psi} - \frac{\dot{V}}{V} \alpha + \frac{1}{Vm} F_{f_s} + \frac{1}{Vm} F_{r_s} + \frac{F_{ady} + F_{gy}}{Vm}, \quad (2.72)$$

$$a_y = \frac{F_{f_s}}{m} + \frac{F_{r_s}}{m} + \frac{F_{ady} + F_{gy}}{m}. \quad (2.73)$$

After subtracting $\frac{F_{gy}}{m}$ from both sides of the last equation we get

$$\underbrace{a_y - \frac{F_{gy}}{m}}_{a_{sy}} = \frac{F_{f_s}}{m} + \frac{F_{r_s}}{m} + \frac{F_{ady}}{m}. \quad (2.74)$$

where a_{sy} is the lateral value measured by the IMU sensor. To understand this imagine that the vehicle is in standstill on a road surface having cross-slope (banking or camber for example). Since the vehicle is in standstill, thus $a_y = 0$, however the lateral acceleration measured by the IMU sensor is $a_y - \frac{F_{gy}}{m} = -\frac{F_{gy}}{m}$. This behavior is a consequence of the working principle of the acceleration sensors, which we have to take into account in this way. Now we convert these equations into matrix form to get closer to the state space representation of our system.

2.9.1 LTI state equation with passive rear wheel steering

Applying equations Eq. (2.71), Eq. (2.72), Eq. (2.49) and Eq. (2.24)) for $\ddot{\psi}$, $\dot{\alpha}$, $\dot{F}_{fs}$ and $\dot{F}_{rs}$ and using Eq. (2.66) for passive wheel steering, we obtain the following LTI state equation of the vehicle model with passive rear wheel steering

$$\underbrace{\begin{bmatrix} \ddot{\psi} \\ \dot{\alpha} \\ \dot{F}_{fs} \\ \dot{F}_{rs} \end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix} 0 & 0 & \frac{l_1-t_f}{J_{zz}} & -\frac{l_2+t_r}{J_{zz}} \\ -1 & -\frac{\dot{V}}{V} & \frac{1}{mV} & \frac{1}{mV} \\ c_y l_1 & c_y V & -\frac{c_y V}{c_f} & 0 \\ -c_y l_2 & c_y V & 0 & -\frac{c_y V}{c_r} \end{bmatrix}}_A \underbrace{\begin{bmatrix} \dot{\psi} \\ \alpha \\ F_{fs} \\ F_{rs} \end{bmatrix}}_x + \underbrace{\begin{bmatrix} 0 \\ 0 \\ -c_y V \\ -c_y V \Xi \end{bmatrix}}_B \underbrace{\delta_f}_u + \underbrace{\begin{bmatrix} 0 & 0 & \frac{1}{J_{zz}} \\ \frac{1}{mV} & \frac{1}{mV} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{W_d} \underbrace{\begin{bmatrix} F_{gy} \\ F_{ady} \\ M_{adz} \end{bmatrix}}_{d_i}. \quad (2.75)$$

2.9.2 LTI state equation with active rear wheel steering

In case of active rear wheel steering, instead of the constant linear relationship between δ_f and δ_r , the rear wheel steering angle δ_r is handled as an independent input. The LTI state space equation of the vehicle model is the following:

$$\underbrace{\begin{bmatrix} \ddot{\psi} \\ \dot{\alpha} \\ \dot{F}_{fs} \\ \dot{F}_{rs} \end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix} 0 & 0 & \frac{l_1-t_f}{J_{zz}} & -\frac{l_2+t_r}{J_{zz}} \\ -1 & -\frac{\dot{V}}{V} & \frac{1}{mV} & \frac{1}{mV} \\ c_y l_1 & c_y V & -\frac{c_y V}{c_f} & 0 \\ -c_y l_2 & c_y V & 0 & -\frac{c_y V}{c_r} \end{bmatrix}}_A \underbrace{\begin{bmatrix} \dot{\psi} \\ \alpha \\ F_{fs} \\ F_{rs} \end{bmatrix}}_x + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ -c_y V & 0 \\ 0 & -c_y V \end{bmatrix}}_B \underbrace{\begin{bmatrix} \delta_f \\ \delta_r \end{bmatrix}}_u + \underbrace{\begin{bmatrix} 0 & 0 & \frac{1}{J_{zz}} \\ \frac{1}{mV} & \frac{1}{mV} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{W_d} \underbrace{\begin{bmatrix} F_{gy} \\ F_{ady} \\ M_{adz} \end{bmatrix}}_{d_i}. \quad (2.76)$$

All states are reachable in both cases.

2.9.3 LTI output equation

Assuming that $\dot{\psi}$, $\ddot{\psi}$, α and a_{sy} are measured, by applying Eq. (2.71) and Eq. (2.74), the output equation in both cases of active and passive rear wheel steering is

$$\underbrace{\begin{bmatrix} \dot{\psi} \\ \ddot{\psi} \\ \alpha \\ a_{sy} \end{bmatrix}}_y = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{l_1 - t_f}{J_{zz}} & -\frac{l_2 + t_r}{J_{zz}} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{m} & \frac{1}{m} \end{bmatrix}}_C \underbrace{\begin{bmatrix} \dot{\psi} \\ \alpha \\ F_{fs} \\ F_{rs} \end{bmatrix}}_x + \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{J_{zz}} \\ 0 & 0 & 0 \\ 0 & \frac{1}{m} & 0 \end{bmatrix}}_{V_d} \underbrace{\begin{bmatrix} F_{gy} \\ F_{ady} \\ M_{adz} \end{bmatrix}}_{d_t}. \quad (2.77)$$

In this output equation all states are observable. These are state space equations of LTI models and the parameters in their matrices are constant at a particular operating point but they may be different at any other ones.

2.10 Augmented plant model

In this part, we augment our plant model with state space equations of additional units: with the EPAS and RASM models, the lateral deviation and heading angle difference, and with an integral action. We will put these equations together into our complete state space model only at the end of this section.

2.10.1 EPAS and RASM models

The EPAS and the RASM are the actuators in our control loop. As we do not know anything about their models, we approximate their dynamics with first order linear systems. This way we identify both actuators at each operating point assuming a first order linear system model. The input signals of these SISO systems are the requested steering angles $\delta_{f_{req}}$ and $\delta_{r_{req}}$ and the output signals are the steering angles δ_f and δ_r . Based on that the transfer function of the EPAS yields

$$G_f(s) = \frac{\delta_f(s)}{\delta_{f_{req}}(s)} = \frac{K_{fe}}{1 + s\tau_{fe}}. \quad (2.78)$$

where the K_{fe} and τ_{fe} parameters are determined by system identification. After rearranging this expression and changing from frequency domain to time domain, we get the following state equation

$$\dot{\delta}_f = -\frac{1}{\tau_{fe}}\delta_f + \frac{K_{fe}}{\tau_{fe}}\delta_{f_{req}}. \quad (2.79)$$

Similarly the transfer function of the RASM yields

$$G_r(s) = \frac{\delta_r(s)}{\delta_{r_{req}}(s)} = \frac{K_{re}}{1 + s\tau_{re}}. \quad (2.80)$$

where the K_{r_e} and τ_{r_e} parameters are determined by system identification. After rearranging this expression and changing from frequency domain to time domain, we get the following state equation

$$\dot{\delta}_r = -\frac{1}{\tau_{r_e}}\delta_r + \frac{K_{r_e}}{\tau_{r_e}}\delta_{r_{req}}. \quad (2.81)$$

We obtain the parameter tuples (K_{f_e}, τ_{f_e}) and (K_{r_e}, τ_{r_e}) by executing measurements, in which we excite the EPAS and RASM systems in each operating point to cover the whole significant frequency range.

The complete plant model is composed of the vehicle model and the steering actuator (EPAS and RASM) models connected to it in series.

2.10.2 Connection to the desired path

In the previous sections we derived the state space equations of the complete plant model, consisting of the model of the steering actuators and that of the vehicle. Now we augment our model with the dynamics that connects our plant to the environment, or more precisely, to the desired path.

In this section we define a dynamical system which characterizes the connection between the vehicle movement and the desired path, with which we augment the state space model in the end. We use two state variables to describe the lateral deviation ν_e and the heading angle difference Θ_e of the vehicle relative to the desired path.

Based on Fig. 2.1, for the time derivative of the lateral deviation at the CoG we obtain

$$\dot{\nu}_e = v_\nu = -|v_t| \sin(\Theta_e - \alpha) \approx V\alpha - V\Theta_e, \quad (2.82)$$

where we used that $\|v_t\| = V$. For the time derivative of the heading angle difference at the CoG we can write that

$$\dot{\Theta}_e = \frac{d}{dt}(\psi_d - \psi) = |v_\xi|\kappa_d - \dot{\psi} = |v_t| \cos(\Theta_e - \alpha)\kappa_d - \dot{\psi} \approx V\kappa_d - \dot{\psi}, \quad (2.83)$$

where ψ_d and κ_d are the yaw angle of the tangent line and the current curvature of the desired path at the CoG of the vehicle. The heading angle of the vehicle is by definition the yaw angle of its longitudinal symmetry line about the Z -axis of the reference frame. At each time step we apply such a coordinate frame whose origin is located at the CoG of the vehicle, and whose axes are parallel with the corresponding axes of the vehicle coordinate frame. We found that with such choice of the reference frame the vehicle heading angle $\psi = 0$, but its rate $\dot{\psi} \neq 0$. The heading angle of the desired path is defined to be the yaw angle ψ_d of the tangent line, drawn to the point of the desired path. The point of tangency on the desired path, through which we draw the tangent line, is designated by the intersection of the desired path-curve and the perpendicular, dropped from the vehicle CoG to the path. Actually, one of our goals is that by reducing the lateral deviation ν_e to zero, the point of tangency and the vehicle CoG coincide. We notice that the above definition of the point of tangency implies that this point moves

together with the vehicle with a tangential speed of v_ξ (see Fig. 2.1). At the point of tangency the curvature of the path-curve is κ_d . We assume that this curvature is online measurable. From this it immediately follows that the time derivative of the tangent line heading angle yields $\dot{\psi}_d = v_\xi \kappa_d$, and that the angle between this tangent line and the longitudinal symmetry line of the vehicle is Θ_e . We note that the lateral deviation and the heading angle difference, and similarly that the curvature of the actual and the desired paths can be measured at an arbitrary point of the vehicle. In the equations Eq. (2.82) and Eq. (2.83) we use the linearized versions of the nonlinear equations, but if the controller performs well, then $\Theta_e - \alpha \approx 0$, so the small angle approximation is considered to be satisfyingly accurate.

2.10.3 Integral action

In real systems it is expected that the steady state error of a simple proportional controller is not zero due to load disturbances and model uncertainties. A frequently used approach to reduce the steady state error to zero is adding integral action to the state feedback. Now we will show in general that we can only achieve asymptotic tracking of as many outputs as many control inputs the system has. Consider the linear system

$$\dot{x} = Ax + Bu, \quad (2.84)$$

$$y = Cx + Du, \quad (2.85)$$

where y is the vector only of those output variables which we want to track asymptotically. Let y_d denote the constant output value of y which is to be tracked asymptotically. Assume that both the steady state vector x and the control input vector u converge to steady state values as $t \rightarrow \infty$, which are denoted by

$$x^* = \lim_{t \rightarrow \infty} x(t), \quad (2.86)$$

$$u^* = \lim_{t \rightarrow \infty} u(t). \quad (2.87)$$

To achieve asymptotic tracking, these x^* and u^* values must satisfy the equation

$$\underbrace{\begin{bmatrix} A & B \\ C & D \end{bmatrix}}_{\Lambda} \begin{bmatrix} x^* \\ u^* \end{bmatrix} = \begin{bmatrix} 0 \\ y_d \end{bmatrix}. \quad (2.88)$$

If we solve this equation for x^* and u^* , we get

$$\begin{bmatrix} x^* \\ u^* \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ y_d \end{bmatrix}. \quad (2.89)$$

The solution is unique if and only if the matrix Λ is non-singular. This means that the matrix must be square, which is fulfilled if the number of rows of matrix C is equal to the number of columns of matrix B . This condition means that asymptotic tracking can only be achieved for as many outputs as many control inputs the system has.

2.10.3.1 Passive rear wheel steering

In case of passive rear wheel steering our plant model has only one control input, namely the requested front wheel steering angle $\delta_{f_{req}}$. Therefore we can asymptotically track only one output, which we choose to be the lateral deviation ν_e , as it is the most reasonable output for our goal of following the desired path. It means that in the integral action we integrate the error of the lateral deviation with respect to time:

$$\epsilon_{\nu_e} = \int (\nu_e - \nu_{ref}) dt. \quad (2.90)$$

By differentiating both sides we get

$$\dot{\epsilon}_{\nu_e} = \nu_e - \nu_{ref} = C_{\nu_e} x - \nu_{ref}, \quad (2.91)$$

where C_{ν_e} is the row of the C matrix of the plant model without integral action that belongs to the plant output ν_e . If we augment our plant model with this state we obtain

$$\begin{bmatrix} \dot{x} \\ \dot{\epsilon}_{\nu_e} \end{bmatrix} = \begin{bmatrix} A & 0 \\ C_{\nu_e} & 0 \end{bmatrix} \begin{bmatrix} x \\ \epsilon_{\nu_e} \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} \delta_{f_{req}} + \begin{bmatrix} 0 \\ -\nu_{ref} \end{bmatrix}, \quad (2.92)$$

where A and B are the original matrices of the plant model without integral actions. The output equation yields

$$\begin{bmatrix} y \\ \epsilon_{\nu_e} \end{bmatrix} = \begin{bmatrix} C & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ \epsilon_{\nu_e} \end{bmatrix}. \quad (2.93)$$

As ϵ_{ν_e} is a reachable state, we can design a control law that drives it to zero in steady state.

2.10.3.2 Active rear wheel steering

In case of active rear wheel steering the plant model has two inputs, the requested front and rear steering angles. Because of that, we can implement two integral actions. We choose the side slip angle α as the second output variable to be asymptotically tracked. Similarly to the integral action of the lateral deviation we can write

$$\dot{\epsilon}_{\alpha_e} = \alpha_e - \alpha_{ref} = C_{\alpha_e} x - \alpha_{ref}, \quad (2.94)$$

where C_{α_e} is the row of the C matrix of the plant model without integral action that belongs to the plant output α . If we augment our plant model with this state we obtain

$$\begin{bmatrix} \dot{x} \\ \dot{\epsilon}_{\nu_e} \\ \dot{\epsilon}_{\alpha_e} \end{bmatrix} = \begin{bmatrix} A & 0 \\ C_{\nu_e} & 0 \\ C_{\alpha_e} & 0 \end{bmatrix} \begin{bmatrix} x \\ \epsilon_{\nu_e} \\ \epsilon_{\alpha_e} \end{bmatrix} + \begin{bmatrix} B & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \delta_{f_{req}} \\ \delta_{r_{req}} \end{bmatrix} + \begin{bmatrix} 0 \\ -\nu_{ref} \\ -\alpha_{ref} \end{bmatrix}, \quad (2.95)$$

where A and B are the original matrices of the plant model without integral actions. The output equation yields

$$\begin{bmatrix} y \\ \epsilon_{\nu_e} \\ \epsilon_{\alpha_e} \end{bmatrix} = \begin{bmatrix} C & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ \epsilon_{\nu_e} \\ \epsilon_{\alpha_e} \end{bmatrix}. \quad (2.96)$$

2.11 State-space representation of the augmented plant model

If we augment the previously defined state space model with the new state space equations derived in the previous section, we get the final LTI state space representation of the plant model.

2.11.1 Passive rear wheel steering

With passive rear wheel steering we obtain

$$\begin{aligned}
 \underbrace{\begin{bmatrix} \ddot{\psi} \\ \dot{\alpha} \\ \dot{F}_{f_s} \\ \dot{F}_{r_s} \\ \dot{\delta}_f \\ \dot{\nu}_e \\ \dot{\Theta}_e \\ \dot{\epsilon}_{\nu_e} \end{bmatrix}}_x &= \underbrace{\begin{bmatrix} 0 & 0 & \frac{l_1-t_f}{J_{zz}} & -\frac{l_2+t_r}{J_{zz}} & 0 & 0 & 0 & 0 \\ -1 & -\frac{\dot{V}}{mV} & \frac{1}{mV} & \frac{1}{mV} & 0 & 0 & 0 & 0 \\ c_y l_1 & c_y V & -\frac{c_y V}{c_f} & 0 & -c_y V & 0 & 0 & 0 \\ -c_y l_2 & c_y V & 0 & -\frac{c_y V}{c_r} & -c_y V \Xi & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{\tau_{fe}} & 0 & 0 & 0 \\ 0 & V & 0 & 0 & 0 & 0 & -V & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} \dot{\psi} \\ \alpha \\ F_{f_s} \\ F_{r_s} \\ \delta_f \\ \nu_e \\ \Theta_e \\ \epsilon_{\nu_e} \end{bmatrix}}_x \\
 &+ \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_B \underbrace{\delta_{f_{req}}}_u + \underbrace{\begin{bmatrix} 0 & 0 & \frac{1}{J_{zz}} & 0 \\ \frac{1}{mV} & \frac{1}{mV} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & V \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{W_d} \underbrace{\begin{bmatrix} F_{g_y} \\ F_{ad_y} \\ M_{ad_z} \\ \kappa_d \end{bmatrix}}_{d_l} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\nu_{ref} \end{bmatrix}}_{W_i}. \tag{2.97}
 \end{aligned}$$

It is clear that if the reference lateral deviation $\nu_{ref} = 0$, then $W_i = 0$. We can easily verify that all states are reachable and observable. Augmenting the output equation as well we get that

$$\underbrace{\begin{bmatrix} \dot{\psi} \\ \ddot{\psi} \\ \alpha \\ a_{s_y} \\ \delta_f \\ \nu_e \\ \Theta_e \\ \epsilon_{\nu_e} \end{bmatrix}}_y = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{l_1-t_f}{J_{zz}} & -\frac{l_2+t_r}{J_{zz}} & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{m} & \frac{1}{m} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_C \underbrace{\begin{bmatrix} \dot{\psi} \\ \alpha \\ F_{f_s} \\ F_{r_s} \\ \delta_f \\ \nu_e \\ \Theta_e \\ \epsilon_{\nu_e} \end{bmatrix}}_x + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{J_{zz}} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \frac{1}{m} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{V_d} \underbrace{\begin{bmatrix} F_{g_y} \\ F_{ad_y} \\ M_{ad_z} \\ \kappa_d \end{bmatrix}}_{d_l} \tag{2.98}$$

2.11.2 Active rear wheel steering

With active rear wheel steering we obtain

$$\underbrace{\begin{bmatrix} \ddot{\psi} \\ \dot{\alpha} \\ \dot{F}_{f_s} \\ \dot{F}_{r_s} \\ \dot{\delta}_f \\ \dot{\delta}_r \\ \dot{\nu}_e \\ \dot{\Theta}_e \\ \dot{\epsilon}_{\nu_e} \\ \dot{\epsilon}_{\alpha_e} \end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix} 0 & 0 & \frac{l_1-t_f}{J_{zz}} & -\frac{l_2+t_r}{J_{zz}} & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & -\frac{\dot{V}}{V} & \frac{1}{mV} & \frac{1}{mV} & 0 & 0 & 0 & 0 & 0 & 0 \\ c_y l_1 & c_y V & -\frac{c_y V}{c_f} & 0 & -c_y V & 0 & 0 & 0 & 0 & 0 \\ -c_y l_2 & c_y V & 0 & -\frac{c_y V}{c_r} & 0 & -c_y V & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{\tau_{fe}} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{\tau_{re}} & 0 & 0 & 0 & 0 \\ 0 & V & 0 & 0 & 0 & 0 & 0 & -V & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} \dot{\psi} \\ \alpha \\ F_{f_s} \\ F_{r_s} \\ \delta_f \\ \delta_r \\ \nu_e \\ \Theta_e \\ \epsilon_{\nu_e} \\ \epsilon_{\alpha_e} \end{bmatrix}}_x + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{K_{fe}}{\tau_{fe}} & 0 \\ 0 & \frac{K_{re}}{\tau_{re}} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}}_B \underbrace{\begin{bmatrix} \delta_{f_{req}} \\ \delta_{r_{req}} \end{bmatrix}}_u + \underbrace{\begin{bmatrix} 0 & 0 & \frac{1}{J_{zz}} & 0 \\ \frac{1}{mV} & \frac{1}{mV} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & V \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{W_d} \underbrace{\begin{bmatrix} F_{gy} \\ F_{ady} \\ M_{adz} \\ \kappa_d \end{bmatrix}}_{d_l} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\nu_{ref} \\ -\alpha_{ref} \end{bmatrix}}_{W_i}. \quad (2.99)$$

Similarly to the plant model with passive steering we can say that if the reference lateral deviation $\nu_{ref} = 0$, and the reference side-slip angle $\alpha_{ref} = 0$, then $W_i = 0$. We can easily verify that all states are reachable and observable. Augmenting the output equation as well we get that

$$\underbrace{\begin{bmatrix} \dot{\psi} \\ \ddot{\psi} \\ \alpha \\ a_{sy} \\ \delta_f \\ \delta_r \\ \nu_e \\ \Theta_e \\ \epsilon_{\nu_e} \\ \epsilon_{\alpha_e} \end{bmatrix}}_y = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{l_1-t_f}{J_{zz}} & -\frac{l_2+t_r}{J_{zz}} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{m} & \frac{1}{m} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_C \underbrace{\begin{bmatrix} \dot{\psi} \\ \alpha \\ F_{f_s} \\ F_{r_s} \\ \delta_f \\ \delta_r \\ \nu_e \\ \Theta_e \\ \epsilon_{\nu_e} \\ \epsilon_{\alpha_e} \end{bmatrix}}_x + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{J_{zz}} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \frac{1}{m} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{V_d} \underbrace{\begin{bmatrix} F_{gy} \\ F_{ady} \\ M_{adz} \\ \kappa_d \end{bmatrix}}_{d_l}. \quad (2.100)$$

3 Longitudinal state space model

Longitudinal control can be handled independently of lateral control with a good approximation. The equilibrium force equation of the CoG written in the longitudinal direction is the following based on Fig. 2.1 and Fig. 2.3 and the derived equations of motion:

$$ma_x = m\dot{V} \cos(\alpha) - mV(\dot{\psi} + \dot{\alpha}) \sin(\alpha) = \sum F_x = F_{f_l} \cos(\delta_f) - F_{f_s} \sin(\delta_f) + F_{r_l} \cos(\delta_r) - F_{r_s} \sin(\delta_r) + F_{ad_x} + F_{g_x} + F_{roll_x} \quad (3.1)$$

where F_{roll_x} is the rolling resistance of the wheels.

Due to its structure, the IMU sensor measures the actual longitudinal acceleration modified by the x -component of the gravitational force due to the road grade, that is

$$a_{x_s} = a_x - g \sin(\lambda), \quad (3.2)$$

where λ is the road inclination angle and $g = -9.81\text{m/s}^2$. The actual longitudinal acceleration can for example be estimated based on the wheel speeds. Let the driving force distribution coefficient between the front and rear wheels be defined as

$$\zeta = \frac{F_{r_l} \cos(\delta_r)}{\underbrace{F_{f_l} \cos(\delta_f) + F_{r_l} \cos(\delta_r)}_{F_x}}, \quad (3.3)$$

where the sum of the lateral forces under the front and rear wheels $F_l = F_{f_l} + F_{r_l}$ is the total driving force under the wheels, F_x is the x -component of F_l . It is evident that if $\delta_f \neq 0$ and $\delta_r \neq 0$, then $F_l \neq F_x$. From this the force distribution between the front and rear wheels in the longitudinal direction can be derived as

$$F_{f_l} = \frac{(1 - \zeta) \cos(\delta_r)}{(1 - \zeta) \cos(\delta_r) + \zeta \cos(\delta_f)} F_l, \quad (3.4)$$

$$F_{r_l} = \frac{\zeta \cos(\delta_f)}{(1 - \zeta) \cos(\delta_r) + \zeta \cos(\delta_f)} F_l. \quad (3.5)$$

Substituting these two equations into Eq. (3.1) for the longitudinal acceleration we get

$$a_x = \frac{F_l \frac{\cos(\delta_f) \cos(\delta_r)}{(1 - \zeta) \cos(\delta_r) + \zeta \cos(\delta_f)} - F_{f_s} \sin(\delta_f) - F_{r_s} \sin(\delta_r) + F_{ad_x} + F_{g_x} + F_{roll_x}}{m}. \quad (3.6)$$

For the dynamics of the powertrain we assume that it can be described by a simple linear first order model

$$F_l = \frac{K_d}{1 + s\tau_d} F_{l_{req}} \quad (3.7)$$

where $F_{l_{req}}$ is the requested driving force that must be developed by the wheels, K_d is the gain and τ_d is the time constant of the powertrain subsystem. Since the input and the output of this subsystem are the same, namely the driving force, we can use that $K_d = 1$.

From the equations above, now we can derive the state equations in matrix form for the longitudinal direction as follows

$$\begin{aligned} \begin{bmatrix} \dot{F}_l \\ \dot{v}_x \end{bmatrix} &= \begin{bmatrix} -\frac{1}{\tau_d} & 0 \\ F_l \frac{\cos(\delta_f) \cos(\delta_r)}{m((1-\zeta) \cos(\delta_r) + \zeta \cos(\delta_f))} & 0 \end{bmatrix} \begin{bmatrix} F_l \\ v_x \end{bmatrix} + \begin{bmatrix} \frac{K_d}{\tau_d} \\ 0 \end{bmatrix} F_{l_{req}} + \\ &+ \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ -\frac{\sin(\delta_f)}{m} & -\frac{\sin(\delta_r)}{m} & \frac{1}{m} & \frac{1}{m} & \frac{1}{m} \end{bmatrix} \underbrace{\begin{bmatrix} F_{f_s} \\ F_{r_s} \\ F_{g_x} \\ F_{ad_x} \\ F_{roll_x} \end{bmatrix}}_{d_l} \end{aligned} \quad (3.8)$$

and the output equation yields

$$\begin{bmatrix} F_l \\ v_x \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} F_l \\ v_x \end{bmatrix}. \quad (3.9)$$

The components of the load disturbance vector d_l are estimated, they are available in every time step. Because of that, we only discuss the rolling resistance F_{roll_x} , which can be separated as

$$F_{f_{roll}} = f_f F_{f_z}, \quad (3.10)$$

$$F_{r_{roll}} = f_r F_{r_z}, \quad (3.11)$$

where the force values F_{f_z} and F_{r_z} are the vertical loads on the front and rear wheels, whose values are depending on the dynamic load transfer due to acceleration and the road elevation angle λ

$$F_{f_z} = F_{f_z}(a_x, \lambda) \quad (3.12)$$

The f_f and f_r values are the rolling resistance factor on the front and rear axle, with a value between $[0.01, 0.02]$, which we can approximate in general with $f_f = f_r = 0.015$.

4 Trajectory generation for the controller algorithm

Based on the data obtained through sensor fusion, we are able to generate the trajectory of the vehicle, which serves as the reference for the control algorithm. Our task is to follow this trajectory using an algorithm that controls the motion of the vehicle. The schematic structure of the control loop [10, 11] with the main parts can be seen in Fig. 4.1.

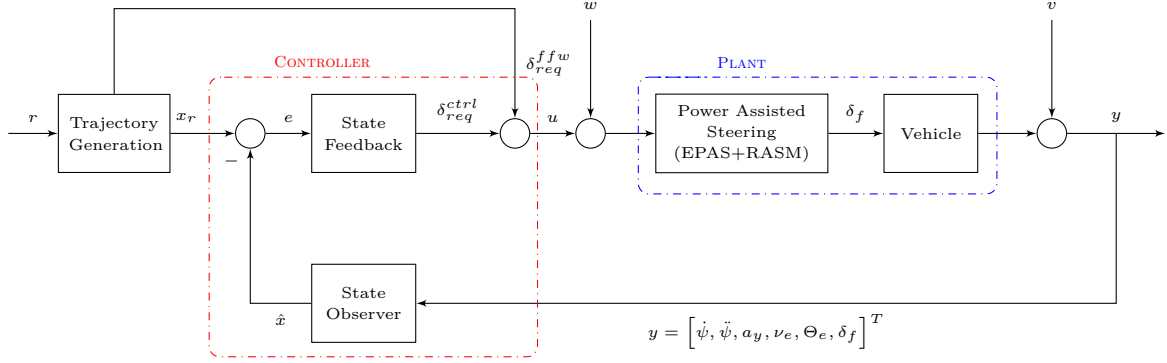


Figure 4.1: Schematic structure of the control loop.

4.1 Trajectory generation

In order to use the controller, it is essential to define the references to be reached. This task is trajectory generation, which consists of the following steps:

1. Path planning: planning a path as a function of spatial coordinates which the vehicle is expected to move along.
2. Motion or trajectory planning: designing the vehicle motion as function of time which then serves as reference input for the controller [12]. In other words we design the trajectory of the inputs, the states, and the outputs of our plant model according to the planned path.

A deeper discussion of this topic is outside the scope of this paper. From the point of view of the controller, the reference trajectory is an input, the planning of which is often separate from the task of controller design. For this reason, we only present the main steps and methods of the trajectory generation without describing the details, which, for those interested, are available in the publications marked as sources.

4.1.1 Path planning

The first phase of the trajectory generation is to design a path along which the vehicle must travel. It seems to be reasonable to design shorter paths and then connect these

shorter paths together to a longer complete maneuver path. Under optimal circumstances the maximum length of a path segment that we can foresee is about 100-150 m, which means that it does not make sense to design longer paths. Specifically we look for the geometric path as function of spatial coordinates in form

$$P(s) = \begin{bmatrix} x(s) \\ y(s) \end{bmatrix}, \quad s : 0 \rightarrow S_f, \quad (4.1)$$

where parameter s is the path length variable, S_f is the length of the complete path. In case of a public road vehicle we can assume that the road is plain and flat, thus x and y coordinates are sufficient.

The path function must be smooth enough that the vehicle can travel along it without too large lateral acceleration values. In practice usually clothoid, sine, cosine, straight and polynomial paths or path segments are used, with which we can avoid the step-function like changes in the reference signals.

4.1.2 Trajectory planning

In the phase of trajectory planning we design the vehicle motion based on the path as function of time, which means that we determine the reference values of the control inputs, the system states and the outputs of the plant subsystem so that the vehicle moves along the planned path. The reference values of the control inputs are used as the feedforward signals, and the reference values of the system states are used as reference signals which are compared to the corresponding measured and/or estimated signals producing the error signal for the controller by this.

4.1.2.1 Differential flatness

It is important that during trajectory generation we need to create a feasible trajectory, which can be realized by the system of the vehicle, EPAS and the controller. A condition for this is that our system needs to be differentially flat [13, 14], which means that we should be able to find a set of outputs (as many as we have inputs) such that all states and inputs can be determined from these outputs without integration. More precisely, the outputs y can be represented as function of the states x and the inputs u and a finite number of its derivatives with respect to time, as well as both the states x and the inputs u can be represented as function of the outputs y and of its derivatives with respect to time, where the components of y are differentially independent. Since our system has only a single input δ_{req} , we need to find only a single flat output. The above definition means that if our system is differentially flat, then by using the single flat output and its derivatives we can write a differential equation system from which we can express both the state $x(t)$ and the input $u(t)$ vectors without integration. Note that the flat outputs do not need to be outputs of the plant model. We can choose any other outputs even any fictitious

outputs. It can also be shown that every controllable linear system is differentially flat. Since our plant model is such a system, it is also differentially flat. Hence our task is to devise an equation system whose unknown variables are the state variables of our plant model. We chose our output to be the quantity $\kappa_d V = (\dot{\psi} + \dot{\alpha})$, where the path curvature κ can either be measured or determined based on the desired path equations, and the vehicle speed V is measured, so our output quantity is also known.

4.1.2.2 Stable Dynamic Inversion

The conditions for the differential flatness actually mean, that we should be able to derive the inverse of the system, with the help of which we can derive the states and the control input for the desired output. From the equations of the state-space model we could derive the system of equations whose solution provides us the reference values of the states x_r and the corresponding reference input u_r . However, this direct method has a huge computational demand, since it leads us to a multivariable, nonlinear system of equations, the solution of which can be problematic in terms of singularities and finding suitable initial conditions for convergence.

Moreover, we have to go without exact inverse because the plant model is not square, that is it does not have the same number of inputs and outputs, as well as it is strictly proper ($D = 0$) or it contains strictly proper transfer functions. Inverse of strictly proper transfer functions are improper and are therefore not realizable. These facts imply that we are not able to derive a perfect inverse plant system. We attempt however to devise a model which approximates the plant inverse fairly accurately.

For this reason, we can use another method, the so called Stable Dynamic Inversion (SDI) for finding a stable approximation with much lower computation effort. This method has the benefit, that we can apply it also for non minimal-phase systems as well as the resulting system is stable. The method has also some drawbacks. One of them is that it may, due to the applied Kalman filter, converge to wrong state values, depending on the initial values. The complete theory is far beyond the scope of this paper, but the topic is explained in more depth in the following literature: [15].

5 Controller design

As we have shown the main steps of generating the reference trajectory in the previous section, according to Fig. 4.1 our next task is to follow this trajectory by implementing an algorithm that controls the motion of the vehicle. The optimal controller is expected to consistently stay within the stability limits under any circumstances, to be robust and to perform sufficiently well. In the case of safety-critical systems (e.g. self-driving vehicles), it is essential that the characteristics of the system can be guaranteed and calculated.

The scientifically based control algorithm is also advantageous from the point of view of cost reduction, since in this case there is no need for long, trial-based control tuning processes. A real vehicle test is only necessary during the system identification, which takes only one day and is much more favorable in terms of resources.

One of the basic conditions of an optimal controller is the good enough knowledge of the mathematical model of the system. In our case, the system to be regulated consists of the vehicle itself, the EPAS and the RASM steering modules.

In this paper, we use a linear parameter-variable (LPV) model with a MPC controller, which works on the entire operating range and which can also take nonlinearities and constraints into account. In this case, however, we would have to solve an online optimization task at each sampling period, which would require a lot of computing capacity. Instead, we define operating points with a given environment, in which we use LTI controllers. Between the operating points we could use an interpolated controller, which means that the corresponding matrices of the controller are interpolated. With this method the entire operating range can be described with a good approximation by using only LTI controllers.

5.1 Model predictive control

Model predictive control (MPC) [16, 17, 18, 19] or receding horizon control (RHC) is an optimal control technique that is designed to optimize the performance of dynamic systems by predicting future system behavior and adjusting control inputs accordingly. The calculated control actions minimize a predefined cost function for a constrained dynamical system over a finite, receding time horizon.

MPC is characterized by its ability to handle complex, nonlinear, and multi-input multi-output (MIMO) systems including time delays and unpredictable disturbances, which makes it suitable for automotive control as well. One of the key advantages is its ability to take system constraints explicitly into consideration, ensuring that the control inputs, outputs and the states remain within specified bounds, which is crucial in safety-critical applications. This predictive nature of MPC allows it to proactively handle disturbances and uncertainties, making it an ideal choice for real-time control of systems with evolving dynamics. Furthermore, MPC is known for its adaptability, as the control strategy can be easily adjusted to meet changing requirements or operational conditions.

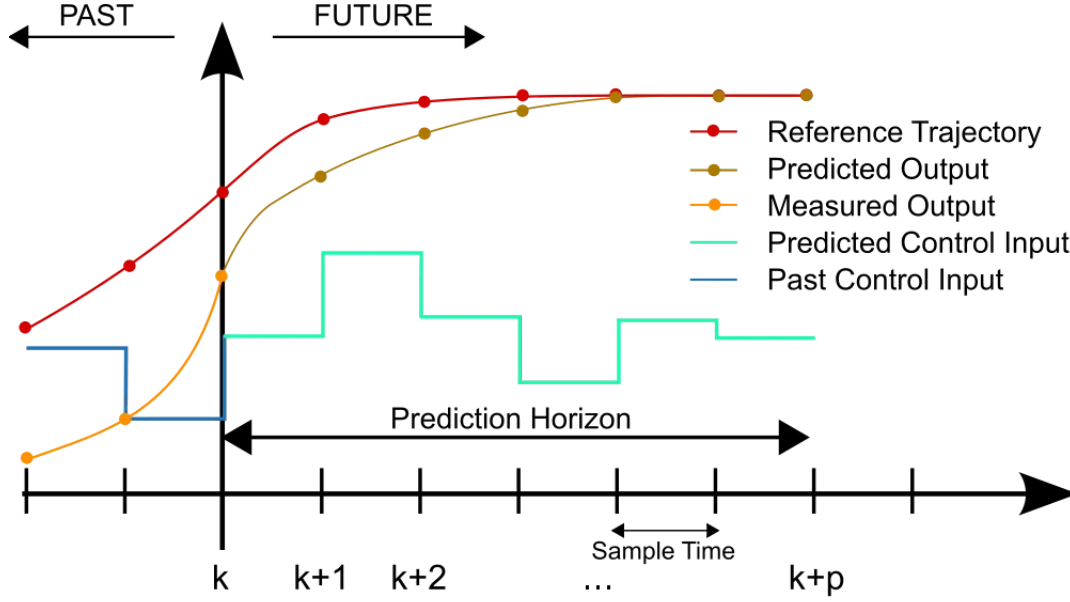


Figure 5.1: The basic working principle of a discrete MPC scheme.
(Figure source: [20])

While MPC offers significant benefits in terms of control performance, it has a high computational demand due to the need for solving online optimization problems at each time step. However, good control performance is essential for solving complex, time-varying problems such as implementing self-driving functions. For this reason, nowadays a lot of emphasis is placed on the research and development of algorithms similar to MPC, which, thanks to the increase in the capacity of processors and the usage of stored operating points, make their use more and more accessible for real world applications.

5.2 Basic MPC control loop

The MPC controller receives the current state of the plant as an input. From that it determines a sequence of control actions that minimizes the cost function over the prediction horizon by solving a constrained optimization problem, which is derived from the internal plant model and the current system states. In the next time step, the controller applies the first control action to the plant, and the optimization is then reiterated for each following time step over a receding horizon.

In case of a linear plant model without constraints by using a quadratic cost function over a horizon which tends to infinity, the MPC is equivalent to linear-quadratic regulator (LQR). In this paper we use a discrete linear time-invariant plant model at every time step with constraints and with a quadratic cost function, and we calculate the control actions over a finite prediction horizon, as shown in Fig. 5.1.

5.3 Control law

In this section we introduce the main steps of the MPC design for linear systems with a quadratic cost function and linear constraints. The main steps of the derivation of the equations were written based on [21]. We consider the following linear, time-invariant model without time-varying output disturbance at every k time step:

$$x_{k+1} = Ax_k + Bu_k, \quad (5.1)$$

$$y_k = Cx_k, \quad (5.2)$$

where $x_k \in \mathbb{R}^n$ is the state vector, $u_k \in \mathbb{R}^m$ is the control input vector, $y_k \in \mathbb{R}^m$ is the output vector.

Let the setpoints or the desired steady-state values for u_k , x_k and y_k be denoted by u_s , x_s and y_s respectively. It means that

$$y_s = Cx_s, \quad (5.3)$$

from which we obtain that

$$u_s = [C(I - A)^{-1}B]^{-1}y_s, \quad (5.4)$$

$$x_s = (I - A)^{-1}Bu_s. \quad (5.5)$$

5.3.1 Cost function

Our task is to find the M -move control sequence $\{u_0, \dots, u_{M-1}\}$, the corresponding state sequence $\{x_0, \dots, x_N\}$ and error sequence $\{e_0, \dots, e_{N-1}\}$, that minimize the finite horizon cost function J defined as

$$\begin{aligned} J_{N,M}(\{x_k\}, \{y_k\}) = & \frac{1}{2}(x_N - x_s)^T P(x_N - x_s) + \frac{1}{2} \sum_{k=0}^{N-1} [(x_k - x_s)^T Q(x_k - x_s)] \\ & + \frac{1}{2} \sum_{k=0}^{M-1} [(u_k - u_s)^T R(u_k - u_s)], \end{aligned} \quad (5.6)$$

where $P \geq 0$, $Q \geq 0$, $R > 0$ are positive semi-definite and positive definite weight matrices, $N \in \mathbb{N}$ is the prediction horizon and $M \in \{m \in \mathbb{N} | m \leq N\}$ is the control horizon. The control input is set to its steady state value after M steps, that is $u_k = u_s$ for all $k \geq M$.

This fixed horizon minimization problem of the cost function is solved at each time step for the current state x_0 . In case of a MPC controller, only the first move of the control sequence is used as the current control input, and then this method is repeated at each time step with a receding horizon.

5.3.2 Quadratic Programming

The cost function Eq. (5.6) of the fixed horizon optimization problem can be transformed into a form that is used in Quadratic Programming (QP) in case of linear constraints on the inputs and outputs. To reach that we write the following set of state equations for the consecutive time steps, using the constraint that $u_k = u_s$ for all $k \geq M$:

$$\begin{aligned}
x_1 &= Ax_0 + Bu_0 \\
x_2 &= Ax_1 + Bu_1 &= A^2x_0 + ABu_0 + Bu_1 \\
&\vdots &\vdots \\
x_M &= Ax_{M-1} + Bu_{M-1} &= A^Mx_0 + A^{M-1}Bu_0 + \cdots + Bu_{M-1} \\
x_{M+1} &= Ax_M + Bu_M &= A^{M+1}x_0 + A^MBu_0 + \cdots + ABu_{M-1} + Bu_s \\
&\vdots &\vdots \\
x_N &= Ax_N + Bu_N &= A^Nx_0 + A^{N-1}Bu_0 + \cdots + A^{N-M}Bu_{M-1} + \sum_{i=0}^{N-M-1} A^iBu_s.
\end{aligned} \tag{5.7}$$

By using the steady-state values of x_s , y_s and u_s , we obtain a similar set of equations for x_s :

$$\begin{aligned}
x_s &= Ax_s + Bu_s \\
x_s &= Ax_s + Bu_s &= A^2x_s + ABu_s + Bu_s \\
&\vdots &\vdots \\
x_s &= Ax_s + Bu_s &= A^Mx_s + A^{M-1}Bu_s + \cdots + Bu_s \\
x_s &= Ax_s + Bu_s &= A^{M+1}x_s + A^MBu_s + \cdots + ABu_s + Bu_s \\
&\vdots &\vdots \\
x_s &= Ax_s + Bu_s &= A^Nx_s + A^{N-1}Bu_s + \cdots + A^{N-M}Bu_s + \sum_{i=0}^{N-M-1} A^iBu_s.
\end{aligned} \tag{5.8}$$

After subtracting the set of equations Eq. (5.8) from Eq. (5.7) we get

$$\begin{aligned}
x_1 - x_s &= A(x_0 - x_s) + B(u_0 - u_s) \\
x_2 - x_s &= A^2(x_0 - x_s) + AB(u_0 - u_s) + B(u_1 - u_s) \\
&\vdots \\
x_M - x_s &= A^M(x_0 - x_s) + A^{M-1}B(u_0 - u_s) + \cdots + B(u_{M-1} - u_s) \\
x_{M+1} - x_s &= A^{M+1}(x_0 - x_s) + A^M B(u_0 - u_s) + \cdots + AB(u_{M-1} - u_s) + \underbrace{B(u_s - u_s)}_{=0} \\
&\vdots \\
x_N - x_s &= A^N(x_0 - x_s) + A^{N-1}B(u_0 - u_s) + \cdots + A^{N-M}B(u_{M-1} - u_s) + \\
&\quad + \underbrace{\sum_{i=0}^{N-M-1} A^i B(u_s - u_s)}_{=0} .
\end{aligned} \tag{5.9}$$

Introducing the following notation

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_M \\ x_{M+1} \\ \vdots \\ x_N \end{bmatrix}, \quad \mathbf{x}_s = \begin{bmatrix} x_s \\ x_s \\ \vdots \\ x_s \\ x_s \\ \vdots \\ x_s \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{M-1} \end{bmatrix}, \quad \mathbf{u}_s = \begin{bmatrix} u_s \\ u_s \\ \vdots \\ u_s \end{bmatrix}, \tag{5.10}$$

where $\mathbf{x}_s$ is an $nN \times 1$ and $\mathbf{u}_s$ is an $mM \times 1$ vector, we can rewrite the state-error evolution equations as the following difference vector

$$\mathbf{x} - \mathbf{x}_s = \Gamma(\mathbf{u} - \mathbf{u}_s) + \Omega(x_0 - x_s) + \underbrace{\Upsilon(\mathbf{u} - \mathbf{u}_s)}_{=0}, \tag{5.11}$$

where

$$\Gamma = \begin{bmatrix} B & 0 & \dots & 0 & 0 \\ AB & B & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ A^{M-1}B & A^{M-2}B & \dots & AB & B \\ A^M B & A^{M-1}B & \dots & A^2 B & AB \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ A^{N-1}B & A^{N-2}B & \dots & \dots & A^{N-M}B \end{bmatrix}, \quad \Omega = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix}$$

$$\Upsilon = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 0 \\ B & 0 & 0 & \dots & 0 & 0 \\ B & AB & 0 & \dots & 0 & 0 \\ B & AB & A^2 B & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ B & AB & A^2 B & \dots & A^{N-M-2}B & 0 \\ B & AB & A^2 B & \dots & A^{N-M-2}B & A^{N-M-1}B \end{bmatrix}. \quad (5.12)$$

To get rid of the summations in the cost function Eq. (5.6), we introduce the following two block diagonal matrices

$$\begin{aligned} \mathbf{Q} &= \text{diag} \{Q, \dots, Q, P\}, \\ \mathbf{R} &= \text{diag} \{R, \dots, R\}. \end{aligned} \quad (5.13)$$

As we have defined the vector notations, we can substitute these expressions to the cost function

$$J_{N,M} = \frac{1}{2}(x_N - x_s)^T P(x_N - x_s) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_s)^T \mathbf{Q}(\mathbf{x} - \mathbf{x}_s) + \frac{1}{2}(\mathbf{u} - \mathbf{u}_s)^T \mathbf{R}(\mathbf{u} - \mathbf{u}_s). \quad (5.14)$$

After substituting Eq. (5.11) into Eq. (5.14) we obtain

$$\begin{aligned} J_{N,M} &= \frac{1}{2}(x_N - x_s)^T P(x_N - x_s) + \frac{1}{2}[\Gamma(\mathbf{u} - \mathbf{u}_s) + \Omega(x - x_s)]^T \mathbf{Q}[\Gamma(\mathbf{u} - \mathbf{u}_s) + \Omega(x - x_s)] \\ &\quad + \frac{1}{2}(\mathbf{u} - \mathbf{u}_s)^T \mathbf{R}(\mathbf{u} - \mathbf{u}_s), \end{aligned}$$

which can be expanded to

$$\begin{aligned} J_{N,M} &= \frac{1}{2}(x_N - x_s)^T P(x_N - x_s) + \frac{1}{2}\mathbf{u}^T \Gamma^T \mathbf{Q} \Gamma \mathbf{u} - \frac{1}{2}\mathbf{u}^T \Gamma^T \mathbf{Q} \Gamma \mathbf{u}_s + \frac{1}{2}\mathbf{u}^T \Gamma^T \mathbf{Q} \Gamma \Omega(x - x_s) - \\ &\quad \frac{1}{2}\mathbf{u}_s^T \Gamma^T \mathbf{Q} \Gamma \mathbf{u} + \frac{1}{2}\mathbf{u}_s^T \Gamma^T \mathbf{Q} \Gamma \mathbf{u}_s - \frac{1}{2}\mathbf{u}_s^T \Gamma^T \mathbf{Q} \Gamma \Omega(x - x_s) + \\ &\quad \frac{1}{2}(x - x_s)^T \Omega^T \mathbf{Q} \Gamma \mathbf{u} - \frac{1}{2}(x - x_s)^T \Omega^T \mathbf{Q} \Gamma \mathbf{u}_s + \frac{1}{2}(x - x_s)^T \Omega^T \mathbf{Q} \Omega(x - x_s) + \\ &\quad \frac{1}{2}\mathbf{u}^T \mathbf{R} \mathbf{u} + \frac{1}{2}\mathbf{u}_s^T \mathbf{R} \mathbf{u}_s - \frac{1}{2}\mathbf{u}^T \mathbf{R} \mathbf{u}_s - \frac{1}{2}\mathbf{u}_s^T \mathbf{R} \mathbf{u}. \end{aligned} \quad (5.15)$$

Now we collect all terms which are independent from u into a common variable $\bar{J}$, as well as factor the variables $\mathbf{u}$, $\mathbf{u}_s$ and $(x - x_s)$

$$\begin{aligned}
J_{N,M} &= \bar{J} + \frac{1}{2} \mathbf{u}^T (\Gamma^T \mathbf{Q} \Gamma + \mathbf{R}) \mathbf{u} - \underbrace{\frac{1}{2} \mathbf{u}^T \Gamma^T \mathbf{Q} \Gamma \mathbf{u}_s - \frac{1}{2} \mathbf{u}_s^T \Gamma^T \mathbf{Q} \Gamma \mathbf{u}}_{-\mathbf{u}^T \Gamma^T \mathbf{Q} \Gamma \mathbf{u}_s} + \\
&\quad \underbrace{\frac{1}{2} \mathbf{u}^T \Gamma^T \mathbf{Q} \Gamma (x - x_s) + \frac{1}{2} (x - x_s)^T \Gamma^T \mathbf{Q} \Gamma \mathbf{u}}_{\mathbf{u}^T \Gamma^T \mathbf{Q} \Gamma (x - x_s)} - \underbrace{\frac{1}{2} \mathbf{u}^T \mathbf{R} \mathbf{u}_s - \frac{1}{2} \mathbf{u}_s^T \mathbf{R} \mathbf{u}}_{-\mathbf{u}^T \mathbf{R} \mathbf{u}_s} = \\
&= \bar{J} + \frac{1}{2} \mathbf{u}^T \underbrace{(\Gamma^T \mathbf{Q} \Gamma + \mathbf{R})}_H \mathbf{u} - \mathbf{u}^T \underbrace{(\Gamma^T \mathbf{Q} \Gamma + \mathbf{R})}_H \mathbf{u}_s + \mathbf{u}^T \underbrace{\Gamma^T \mathbf{Q} \Gamma (x - x_s)}_F = \\
&= \bar{J} + \frac{1}{2} \mathbf{u}^T H \mathbf{u} + \mathbf{u}^T [F(x - x_s) - H \mathbf{u}_s],
\end{aligned} \tag{5.16}$$

where the introduced notations represent

$$H = \Gamma^T \mathbf{Q} \Gamma + \mathbf{R}, \quad F = \Gamma^T \mathbf{Q} \Gamma. \tag{5.17}$$

The last substitution into the cost function can be considered as the elimination of the equality constraints of the state equations Eq. (5.1)-Eq. (5.3).

Now we search for $\mathbf{u}$, that minimizes the above cost function. Based on the more detailed examination of the terms of the cost function in [21] we can use the following statements:

- $J_{N,M} = J_{N,M}(\mathbf{u}) : S \rightarrow \mathbb{R}$, where S is a complex set.
- If H , the so called Hessian matrix, is positive definite, then the function $J_{N,M}(\mathbf{u})$ is strictly convex and $\bar{\mathbf{u}}$ is unique global minimum.
- If H is symmetric and invertible, then $\bar{\mathbf{u}} = -H^{-1}[F(x - x_s) - H \mathbf{u}_s]$.

In the following section we will prove that the Hessian matrix

$$H = \Gamma^T \mathbf{Q} \Gamma + \mathbf{R} \tag{5.18}$$

is positive definite, from which we can deduce the minimum of the cost function.

The $\mathbf{Q}$ term is a block diagonal matrix with the matrices Q and P in the diagonal. As declared in the definition of Eq. (5.6), $P \geq 0$ and $Q \geq 0$, that is they are positive semi-definite matrices. As a consequence the resulting $\mathbf{Q}$ matrix is also positive semi-definite. The matrix $\mathbf{R}$ is also a block diagonal matrix with matrices R in its diagonal. As we also declared in Eq. (5.6) it is positive definite, that is $R > 0$. This implies that not only R but also $\mathbf{R}$ is positive definite. We show that if $\mathbf{Q}$ is positive semi-definite then the matrix $\Gamma^T \mathbf{Q} \Gamma$ is positive semi-definite and from that it follows that H is positive definite. Applying the quadratic form $v^T H v$ for H , we get that

$$v^T (\Gamma^T \mathbf{Q} \Gamma + \mathbf{R}) v = v^T (\Gamma^T \mathbf{Q} \Gamma) v + v^T \mathbf{R} v = \underbrace{(\Gamma v)^T}_{\omega^T} \underbrace{\mathbf{Q}}_{\omega} (\Gamma v) + \underbrace{v^T \mathbf{R} v}_{\geq 0} = \underbrace{\omega^T \mathbf{Q} \omega}_{\geq 0} + \underbrace{v^T \mathbf{R} v}_{> 0} > 0, \tag{5.19}$$

where $\mathbf{Q} \in \mathbb{R}^{nN \times nN}$, $\Gamma \in \mathbb{R}^{nN \times mM}$, $\mathbf{v} \in \mathbb{R}^{mM}$, $\mathbf{v} \neq 0$ and $\omega \in \mathbb{R}^{mM}$. This means that $\Gamma^T \mathbf{Q} \Gamma$ is positive semi-definite based on the definition of positive semi-definite matrices. The property is known that the sum of a positive semi-definite matrix and a positive definite matrix is always positive definite, from which it follows that H is positive definite. This implies that $\bar{\mathbf{u}}$ is a unique global minimum. Since H is positive definite, thus it is symmetric. Therefore the derivative of $J_{N,M}(\mathbf{u})$ is

$$\begin{aligned} \frac{\partial J_{N,M}(\mathbf{u})}{\partial \mathbf{u}} &= \frac{\partial \left\{ \bar{J} + \frac{1}{2} \mathbf{u}^T H \mathbf{u} + \mathbf{u}^T [F(x - x_s) - H \mathbf{u}_s] \right\}}{\partial \mathbf{u}} = \\ \mathbf{u}^T H + [F(x - x_s) - H \mathbf{u}_s] &= H \mathbf{u} + [F(x - x_s) - H \mathbf{u}_s] = 0. \end{aligned} \quad (5.20)$$

Solving the equation for $\mathbf{u}$ we obtain the global minimum $\bar{\mathbf{u}}$

$$\bar{\mathbf{u}} = -H^{-1} [F(x - x_s) - H \mathbf{u}_s] = -H^{-1} F(x - x_s) + \mathbf{u}_s. \quad (5.21)$$

If H is positive definite then its eigenvalues are positive, so 0 is not an eigenvalue. This means that the system of equations $H \mathbf{u}$ has only the trivial $\mathbf{u} = \mathbf{0}$ solution, and so H is invertible. We examine this statement a bit further. By definition of the eigenvalues we can write the above system of equations as

$$H \mathbf{u} = \lambda \mathbf{u} = \mathbf{0}, \quad (5.22)$$

for some eigenvector $\mathbf{u} \neq \mathbf{0}$ the solution would be $\lambda = 0$, which contradicts the fact that 0 is not an eigenvalue of H . That is the equation above has only trivial solution. In other words the null space of the H matrix is the null-vector, that is $\text{Null}(H) = \mathbf{0}$, which implies that H is invertible.

The controller deploys only the first element of $\bar{\mathbf{u}}$ (first m components of the expression in Eq. (5.21)), which has a linear time-invariant feedback structure of the form

$$\bar{u} = -K(x - x_s) + u_s, \quad (5.23)$$

where K is defined as the first m rows of the matrix $H^{-1}F$. The control law Eq. (5.23) is the control used by the MPC algorithm if the problem is unconstrained.

5.4 Constraint handling

One of the main advantages of MPC control is its ability of handling constraints. In general we can introduce magnitude and rate inequality constraints on the plant input and output, which can be expressed as follows:

$$\begin{aligned} u_{\min} &\leq u_k \leq u_{\max}, \quad k = 0, 1, \dots, M-1, \\ y_{\min} &\leq y_k \leq y_{\max}, \quad k = 0, 1, \dots, N-1, \\ \delta u_{\min} &\leq u_k - u_{k-1} \leq \delta u_{\max}, \quad k = 0, 1, \dots, M-1, \end{aligned} \quad (5.24)$$

where u_{-1} is the input used in the previous step of the receding horizon implementation, which is stored for the use of the current fixed horizon optimization. It is also

a requirement that the output setpoint y_s and the corresponding input and state setpoints u_s and x_s must be feasible, which means they also satisfy the required constraints: $u_{min} \leq u_s \leq u_{max}$ and $y_{min} \leq y_s \leq y_{max}$.

We can define constraints for the states as well in the form

$$x_k \in \mathbb{X}_k \quad \text{for } k = 1, 2, \dots, N, \quad (5.25)$$

where $\mathbb{X}_k$ is a polyhedral set of the form

$$\mathbb{X}_k = \left\{ x \in \mathbb{R}^n : W_{L_k}^x \leq L_k x \leq W_{H_k}^x \right\}, \quad (5.26)$$

or

$$\mathbb{X}_k = \left\{ x \in \mathbb{R}^n : L_k x \leq W_k^x = \begin{bmatrix} W_{H_k}^x \\ -W_{L_k}^x \end{bmatrix} \right\}, \quad (5.27)$$

where $W_{L_k}^x$ and $W_{H_k}^x$ are the lower and upper state constraint vectors respectively, and W_k^x state constraint vector is the combination of these two at the k^{th} prediction step. The setpoint state vector x_s must satisfy these constraints as well, that is $x_s \in \mathbb{X}_k$ for each k . The input, output and state constraints can be written as linear constraints on $\mathbf{u}$ of the form

$$W_L \leq L\mathbf{u} \leq W_H, \quad (5.28)$$

or after rearrangement

$$\underbrace{\begin{bmatrix} L \\ -L \end{bmatrix}}_{L^*} \mathbf{u} \leq \underbrace{\begin{bmatrix} W_H \\ -W_L \end{bmatrix}}_W, \quad (5.29)$$

$$L^* \mathbf{u} \leq W. \quad (5.30)$$

These two forms of the linear constraints are equivalent, which one to use depends on the solver we intend to use. For instance the QP solver `quadprog` function of MATLAB uses the second form Eq. (5.30), but the `osqp` solver which we implemented in our model uses the first form Eq. (5.28). As the derivation above shows, the transformation from one form into the other is practically a rearrangement of the inequality.

In the following we introduce how to compute the matrices of the linear constraints for the first form Eq. (5.28), where

$$W_L = \begin{bmatrix} \mathbf{u}_{min} \\ \mathbf{y}_{min} \\ \delta \mathbf{u}_{min} \\ \tilde{W}_L \end{bmatrix}, \quad L = \begin{bmatrix} I_{Mm} \\ \Psi \\ E \\ \tilde{L} \end{bmatrix}, \quad W_H = \begin{bmatrix} \mathbf{u}_{max} \\ \mathbf{y}_{max} \\ \delta \mathbf{u}_{max} \\ \tilde{W}_H \end{bmatrix}. \quad (5.31)$$

The structure of the individual matrices inside the L , W_L and W_H matrices is the following

$$\mathbf{u}_{min} = \begin{bmatrix} u_{min} \\ \vdots \\ u_{min} \end{bmatrix}, \mathbf{u}_{max} = \begin{bmatrix} u_{max} \\ \vdots \\ u_{max} \end{bmatrix}, \delta\mathbf{u}_{min} = \begin{bmatrix} u_{-1} + \delta u_{min} \\ \delta u_{min} \\ \vdots \\ \delta u_{min} \end{bmatrix}, \delta\mathbf{u}_{max} = \begin{bmatrix} u_{-1} + \delta u_{max} \\ \delta u_{max} \\ \vdots \\ \delta u_{max} \end{bmatrix}, \quad (5.32)$$

$$\mathbf{y}_{min} = \begin{bmatrix} y_{min} - CAx \\ y_{min} - CA^2x \\ \vdots \\ y_{min} - CA^Mx \\ y_{min} - CA^{M+1}x - CBu_s \\ \vdots \\ y_{min} - CA^{N-1}x - \sum_{i=0}^{N-M-2} CA^i Bu_s \end{bmatrix}, \quad (5.33)$$

$$\mathbf{y}_{max} = \begin{bmatrix} y_{max} - CAx \\ y_{max} - CA^2x \\ \vdots \\ y_{max} - CA^Mx \\ y_{max} - CA^{M+1}x - CBu_s \\ \vdots \\ y_{max} - CA^{N-1}x - \sum_{i=0}^{N-M-2} CA^i Bu_s \end{bmatrix}, \quad (5.34)$$

where x is the initial state, u_{max} , u_{min} , δu_{max} , δu_{min} , y_{max} and y_{min} are the vectors of constraint limits defined in Eq. (5.24).

I_{Mm} is $Mm \times Mm$ identity matrix; Ψ is the following $m(N-1) \times mM$ matrix

$$\Psi = \begin{bmatrix} CB & 0 & \dots & 0 & 0 \\ CAB & CB & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ CA^{M-1}B & CA^{M-2}B & \dots & CAB & CB \\ CA^M B & CA^{M-1}B & \dots & CA^2B & CAB \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ CA^{N-2}B & CA^{N-3}B & \dots & CA^{N-M}B & CA^{N-M-1}B \end{bmatrix}, \quad (5.35)$$

E is the following $mM \times mM$ matrix

$$E = \begin{bmatrix} I_m & 0 & \dots & 0 \\ -I_m & I_m & \dots & 0 \\ & \ddots & \ddots & \\ 0 & \dots & -I_m & I_m \end{bmatrix}, \quad (5.36)$$

where I_m is $m \times m$ identity matrix.

Now we introduce

$$\mathbf{L} = \text{diag}\{L_1, L_2, \dots, L_N\}, \quad (5.37)$$

where $L_1, L_2, \dots, L_N$ are the state constraint matrices given in Eq. (5.27).

Using Eq. (5.11) we can write

$$\mathbf{x} = \Gamma \mathbf{u} + \Omega x_0 + \Upsilon \mathbf{u}_s. \quad (5.38)$$

Substituting this expression into the state constraint $\mathbf{W}_L^x \leq \mathbf{Lx} \leq \mathbf{W}_H^x$, where

$$\mathbf{W}_L^x = \begin{bmatrix} W_{L_1}^x \\ \vdots \\ W_{L_N}^x \end{bmatrix}, \quad \mathbf{W}_H^x = \begin{bmatrix} W_{H_1}^x \\ \vdots \\ W_{H_N}^x \end{bmatrix}, \quad (5.39)$$

we obtain

$$\mathbf{W}_L^x \leq \mathbf{L}\Gamma \mathbf{u} + \mathbf{L}\Omega x_0 + \mathbf{L}\Upsilon \mathbf{u}_s \leq \mathbf{W}_H^x, \quad (5.40)$$

after rearrangement

$$\underbrace{\mathbf{W}_L^x - \mathbf{L}\Omega x_0 - \mathbf{L}\Upsilon \mathbf{u}_s}_{\tilde{\mathbf{W}}_L} \leq \underbrace{\mathbf{L}\Gamma}_{\tilde{\mathbf{L}}} \mathbf{u} \leq \underbrace{\mathbf{W}_H^x - \mathbf{L}\Omega x_0 - \mathbf{L}\Upsilon \mathbf{u}_s}_{\tilde{\mathbf{W}}_H}, \quad (5.41)$$

that is

$$\tilde{\mathbf{W}}_L \leq \tilde{\mathbf{L}} \mathbf{u} \leq \tilde{\mathbf{W}}_H. \quad (5.42)$$

In this way, the constraints on the states were also brought into a form suitable for the QP solver. The elements of the resulting $\tilde{\mathbf{W}}_L$ and $\tilde{\mathbf{W}}_H$ matrices are as follows

$$\tilde{\mathbf{W}}_L = \begin{bmatrix} W_{L_1}^x \\ \vdots \\ W_{L_N}^x \end{bmatrix} - \text{diag}\{L_1, L_2, \dots, L_N\} \begin{bmatrix} Ax \\ \vdots \\ A^M x \\ A^{M+1}x + Bu_s \\ A^{M+2}x + ABu_s + Bu_s \\ \vdots \\ A^N x + \sum_{i=0}^{N-M-1} A^i Bu_s \end{bmatrix}, \quad (5.43)$$

$$\tilde{\mathbf{W}}_H = \begin{bmatrix} W_{H_1}^x \\ \vdots \\ W_{H_N}^x \end{bmatrix} - \text{diag}\{L_1, L_2, \dots, L_N\} \begin{bmatrix} Ax \\ \vdots \\ A^M x \\ A^{M+1}x + Bu_s \\ A^{M+2}x + ABu_s + Bu_s \\ \vdots \\ A^N x + \sum_{i=0}^{N-M-1} A^i Bu_s \end{bmatrix}, \quad (5.44)$$

where $x = x_0$ is the initial state, $L_1, L_2, \dots, L_N, W_{L_1}^x, W_{L_2}^x, \dots, W_{L_N}^x, W_{H_1}^x, W_{H_2}^x, \dots, W_{H_N}^x$ are the state constraint matrices given in the polyhedral set Eq. (5.27) and Γ, Ω and Υ are given in Eq. (5.12).

5.4.1 The constrained QP problem

Our problem now is minimizing the cost function Eq. (5.6) subject to the inequality constraints Eq. (5.24)-Eq. (5.27), which can be formulated as a QP problem of minimizing Eq. (5.16) subject to Eq. (5.28), that is

$$\begin{aligned} \min_{\mathbf{u}} \quad & \frac{1}{2} \mathbf{u}^T H \mathbf{u} + \mathbf{u}^T [F(x - x_s) - H \mathbf{u}_s], \\ \text{subject to:} \quad & \\ & W_L \leq L^* \mathbf{u} \leq W_H. \end{aligned} \tag{5.45}$$

As $\bar{J}$ in Eq. (5.16) is independent of $\mathbf{u}$, it is not included in Eq. (5.45).

The optimal solution $\bar{\mathbf{u}}(x)$ to the QP Eq. (5.45) yields:

$$\bar{\mathbf{u}}(x) = \arg \min_{W_L \leq L^* \mathbf{u} \leq W_H} \frac{1}{2} \mathbf{u}^T H \mathbf{u} + \mathbf{u}^T [F(x - x_s) - H \mathbf{u}_s]. \tag{5.46}$$

As we have shown in the previous section, H is a positive definite, which implies that the QP problem is convex. There are standard numerical procedures for solving such optimization problems, which are called QP algorithms, details of which are beyond the scope of this paper.

Once the QP problem Eq. (5.45) is solved, we can apply the receding horizon algorithm. At the current time step k only the first control move is executed, which is formed by the first m components of the optimal solution vector $\bar{\mathbf{u}}(x)$ in Eq. (5.46). From this we obtain the control law for every time step of the form

$$u_k = \mathcal{K}(x_k, u_s, y_s) \tag{5.47}$$

where $x_k = x$ is the current state, u_s and y_s are the steady-state values of the control input and output values, which are considered in the optimization problem Eq. (5.46). After that, the entire procedure is repeated at the next time instant, with the optimization horizon kept constant.

5.5 Control parameters

The main controller parameters of the MPC algorithm are the sampling time T_s , the prediction horizon N , the control horizon M , the constraints and the weight matrices. The set and tuning of these parameters are beyond the scope of this study. However, good rules of thumb and tuning methods (e.g. Bryson's rule, output weighting) are introduced in the literature of MPC, for example in [22].

6 Numerical simulations

To implement the previously derived equations for vehicle motion control including the state-space equations of the model, the trajectory generation and the control algorithm we used Simulink, which is a MATLAB-based, block diagram programming environment for Model-Based Design.

To simulate and analyze the dynamical system characteristics, we connected the Simulink model to the CarMaker simulation software. CarMaker is an automotive simulation software which provides a test platform to implement virtual test scenarios for the application areas of autonomous vehicles, ADAS, powertrain and vehicle dynamics. With the help of this software, we could test the operation of our controller integrated into a vehicle on different, customizable tracks and maneuvers. This was possible by replacing the original vehicle motion control part in the CarMaker Simulink model with our own model, so we could test our own controller in the simulation environment. At the same time, we were able to realistically visualize the motion of the vehicle as it can be seen on Fig. 6.1, based on which the behavior of the controller can be viewed and analyzed.

During the simulations, the inputs, outputs and state variables of our model can be continuously displayed and plotted. To perform the simulations, we used the model with active rear wheel steering. In the following, we present the results of the numerical simulations for some individually defined test scenarios, and based on them we compare and evaluate the performance of the controller.

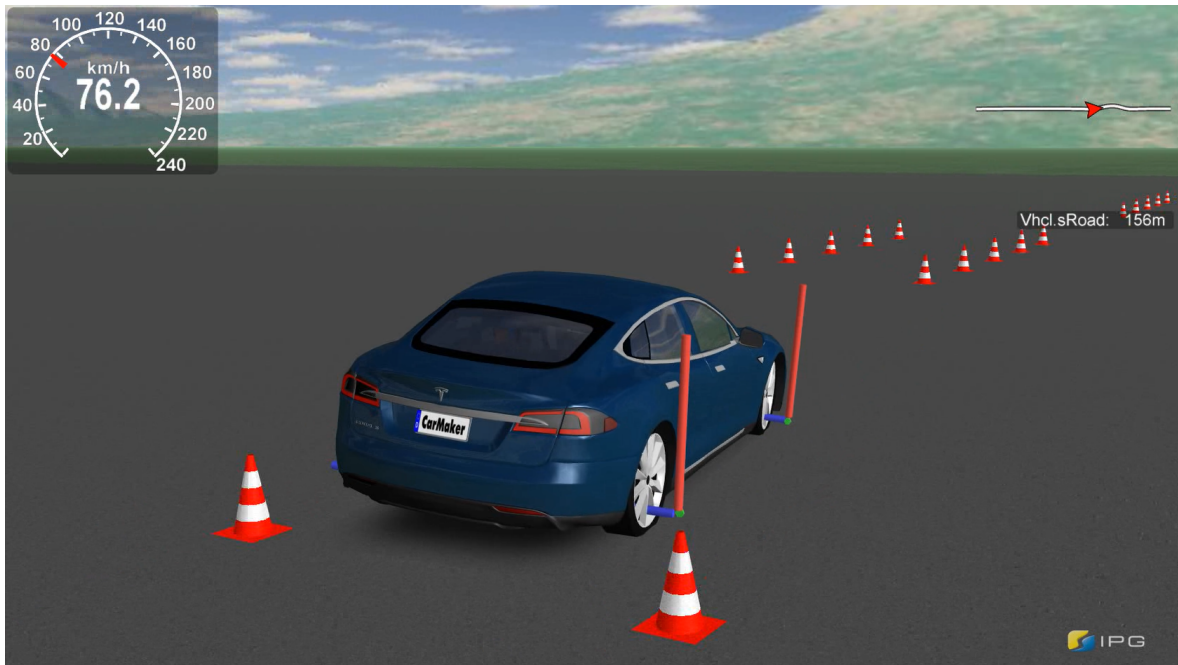


Figure 6.1: Maneuvering a model car in the CarMaker simulation environment.

6.1 Simple U-turn test

In the first test, we drove a car along a track including a U-turn with a radius of 150 meters at a speed of 90 km/h, under normal road conditions. Although in reality in the EU the minimum curvature radius on public roads is about 300 meters for the 90 km/h speed limit, based on the state variables our controller performed excellently even on this track, which can be seen on Fig. 6.2.

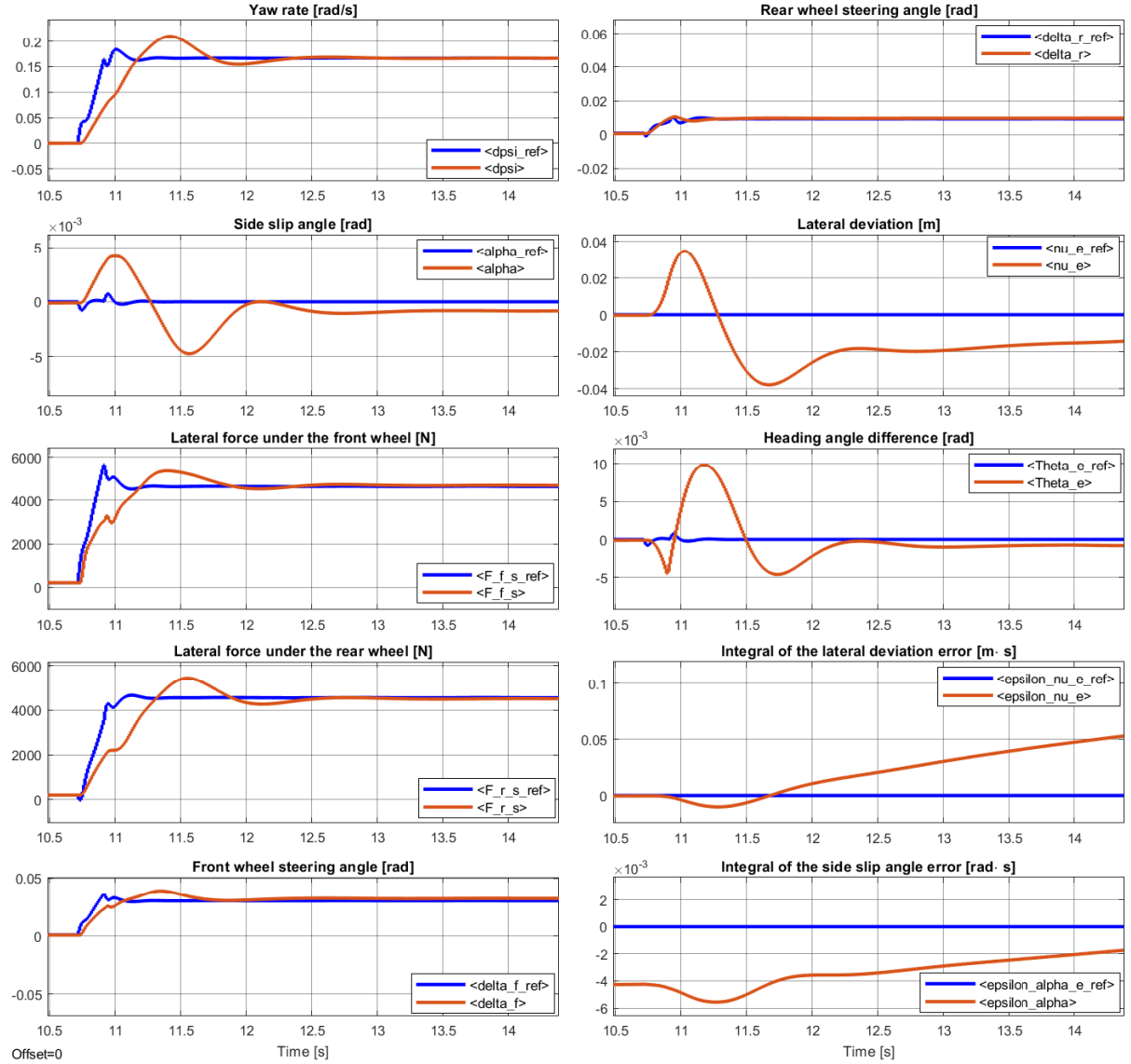


Figure 6.2: Time function of the state variables in case of the U-turn test.

The maximum lateral deviation was 3.8 cm at the beginning of the turn, which would be even smaller in the case of roads with clothoid geometry, which are used in reality. As we have defined a circular path with constant radius after a straight section, the curvature of the track is not continuous. On the straight section, the curvature of the track is infinite, and in the turn it is constant 150 meters. The discontinuous transition between the two generates Heaviside step-like reference signals. After the small overshoot

resulting from this, the lateral deviation is reduced to 2 cm, and thanks to the integral action, it is reduced even further along the turn. That small deviation value can be considered perfectly applicable from a practical point of view, so this test proves the proper functioning of our controller.

6.2 Moose test

A frequently used test in the automotive industry is the so called Moose test standardized in ISO 3888-2 [23], which is an evasive maneuver test that examines the stability and maneuverability of vehicles. During the test, the vehicle must perform two lane changes within a short distance, which is intended to demonstrate the avoidance of an obstacle that suddenly appears on the road. The standard precisely defines the conditions of the test and the position of the track edge marking cones, which can be seen on Fig. 6.3. The test is carried out several times in a row at increasing speeds, with the goal of finding the stability limit, when the vehicle is no longer able to stay on the track without skidding or knocking out the cones.

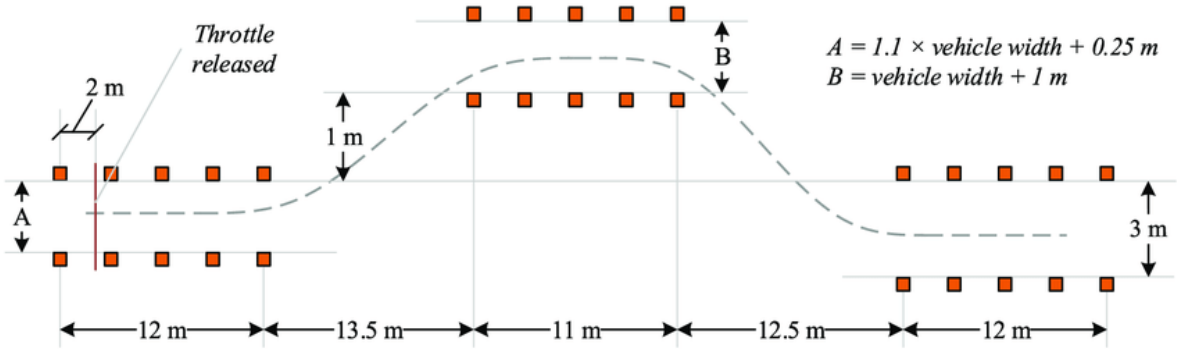


Figure 6.3: The double lane-change maneuver according to ISO-3888-2.
(Figure source: [24])

For the simulations, we created a test track in the CarMaker software in accordance with the standard, and on this we tested the performance of our controller at different speeds.

6.2.1 Passing the test with an entrance speed of $v = 70$ km/h

At an entry speed of 70 km/h, our vehicle was still able to perform the maneuver without hitting the cones. As shown in Fig. 6.4, our controller was able to correctly track the reference signals, our system remained stable through the whole process. The maximum side slip angle was around 7.5° , the maximum lateral deviation was 21 cm, and the maximum heading angle difference was about 9° .

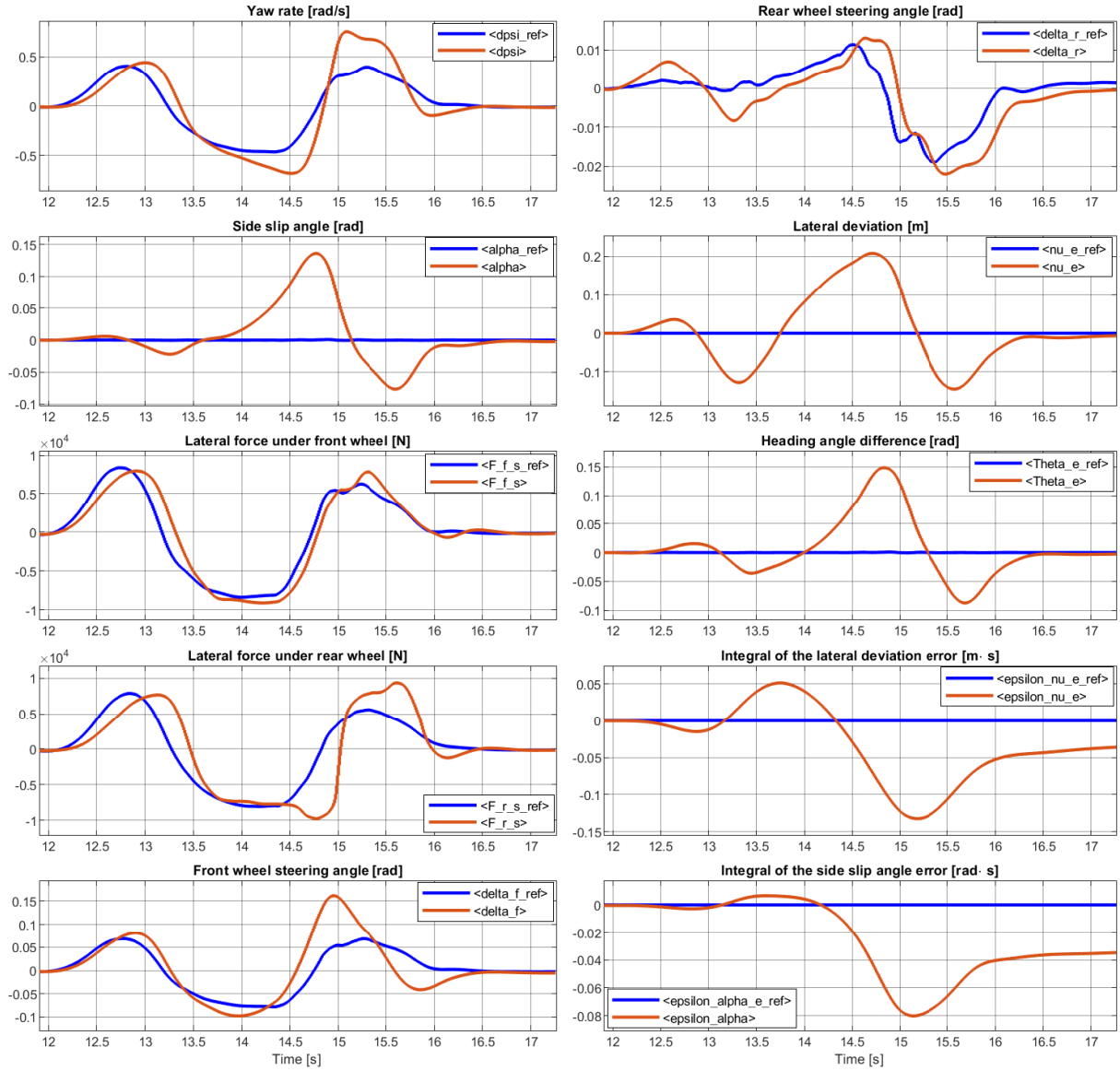


Figure 6.4: Time function of the state variables in case of $v = 70$ km/h entrance speed.

6.2.2 Failing the test with an entrance speed of $v = 77$ km/h

At an entry speed of 77 km/h, however, our vehicle was no longer able to perform the maneuver without hitting the cones. The controller remained stable and the car returned to the reference track after a small oscillation, but it could not stay within the limits specified in the standard. It can be seen by the time function of the state variables indicating the deviations from the reference in Fig. 6.5. The maximum side slip angle was around 4.6° , the maximum lateral deviation increased significantly to more than 2 meters, and the maximum heading angle difference reached 15° . These results are no longer acceptable for the test.

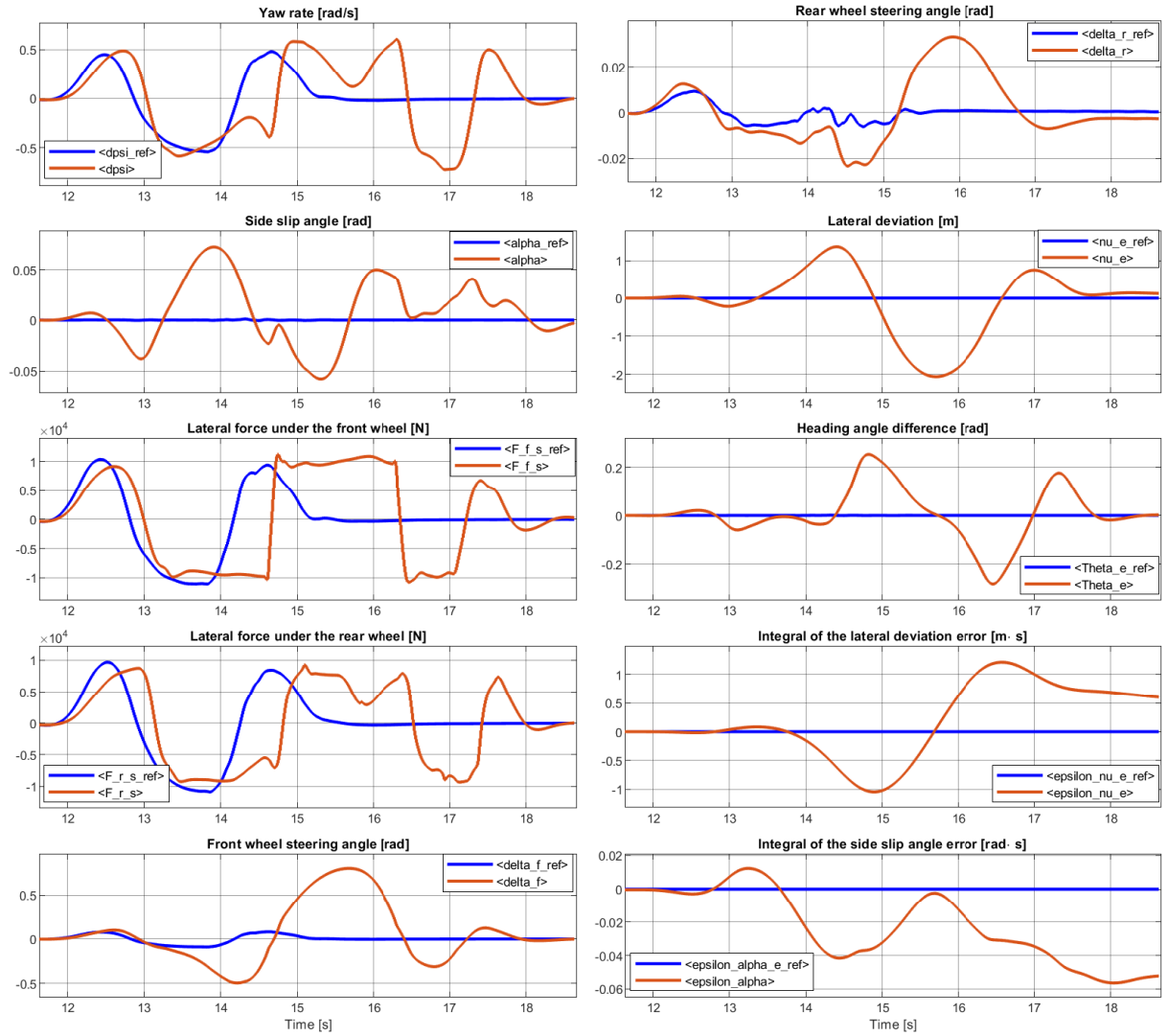


Figure 6.5: Time function of the state variables in case of $v = 77$ km/h entrance speed.

6.2.3 Stability loss with an entrance speed of $v = 82$ km/h

Based on the moose test videos available on the Internet, a car similar to the vehicle we used in the simulation was able to successfully complete the test at a maximum speed of 82 km/h in reality. For this reason, we also run the simulation at this speed. Based on the results summarized in Fig. 6.6 and the CarMaker animation, it can be clearly seen that the vehicle completely lost its stability during the maneuver, the car even spun around its axis.

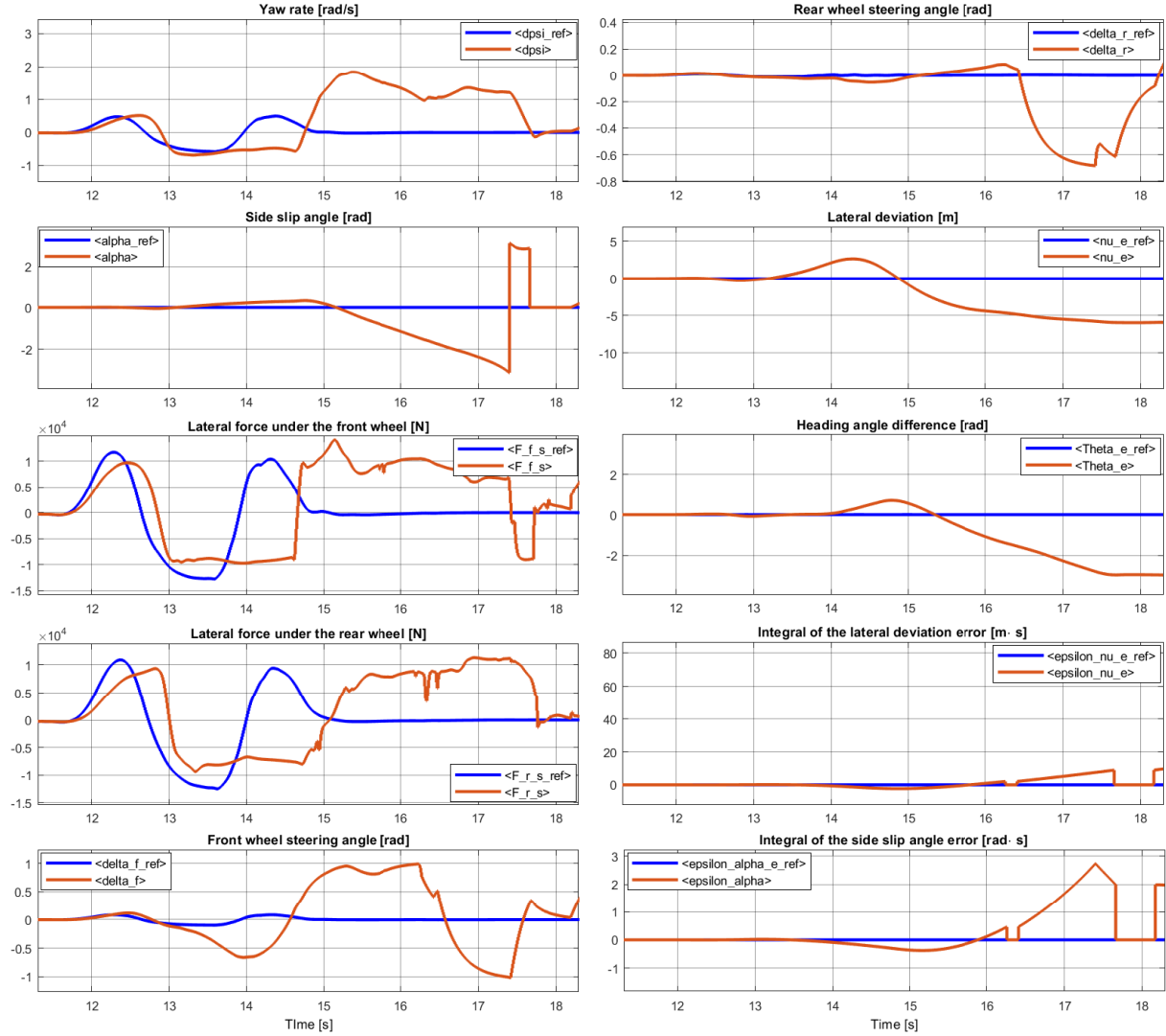


Figure 6.6: Time function of the state variables in case of $v = 82$ km/h entrance speed.

6.3 Evaluation of the results

Based on the results of the moose test simulations looking for the stability limit, the maximum speed was roughly 70 km/h, with which our controller was able to successfully follow the reference trajectory and thus passed the test. Most cars can perform this test

between 65-75 km/h, 72 km/h is needed to pass the test, and the best result so far has been 85 km/h [25]. In the videos showing the test on different cars, it is clear that these maneuvers are really tried to the last centimeter by professional drivers, which in our case could not be realized in the simulation, but it was not our goal either. With this in mind, the 70 km/h result we achieved can be considered good, and the test can also be suitable for further analysis from the point of view of examining the performance of the controller.

On the other hand, the simple U-turn test showed that our controller works very precisely under normal driving conditions. Since our model and self-driving functions in general are both designed for normal driving conditions, tracks like this simpler test are more relevant and representative of the controller performance. Nevertheless, testing the behavior around the stability limit is also essential in such a safety-critical application.

7 Summary

The actuality of the topic of autonomous driving is unquestionable, and as can already be seen in the vehicles used on a daily basis, different levels of self-driving will become more and more common in vehicles. The increase in computing capacities makes it possible to achieve better and better perception results and controller performance, although the electronic control units used in the automotive industry are still far behind the state-of-the-art technology. For this reason the real application of MPC in vehicle motion control is yet to be seen due to its high computational demand. Yet it is understandable that a lot of emphasis is placed on MPC in vehicle motion control developments these days, since its predictive nature is excellent for applications related to self-driving.

7.1 Results

In summary, it can be said that achieving full control of a self-driving car is a very complex task. Each smaller subtask contains a lot of challenges and optimization opportunities, which can even be critical from the point of view of final applicability.

Nevertheless, the methods presented in the study proved to be well applicable, as shown by the simulation results. The applicability of our vehicle model could be tested during real system identification, and additional simulations can be run in the simulation environment to further tune, optimize and test the performance of the MPC controller. Overall, the paper provides a good basis for presenting the basic steps for designing self-driving functions and facilitates further exploration of specific areas of the topic.

7.2 Conclusions, outlook

In reality, due to the time delay of the sensors and the steering actuators and the time required for calculations and communication, a non-negligible time delay (up to hundreds of milliseconds) appears in the control loop, which can also lead to instabilities [26, 27]. Giving recommendations to compensate the effect of this time delay on stability and considering it in the simulation environment is a next task in order to model the real system even more accurately.

Among the further development possibilities are the examination of nonlinear models [28, 29, 30] and their effect on the stability of the controller, the connection of the lateral and longitudinal controller, the implementation of a derivative action and connecting the path generated by perception in real time with the controller algorithm. A more in-depth examination of the performance, stability and robustness of the controller is also essential in order to achieve a reliable, applicable final result.

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